

Inferential Uncertainty in RNA-Seq and its downstream applications

Recap

In the last lecture we covered:

- RNA-Seq quantification problem
- Generative model for RNA-Seq
- (Range-factorized) Equivalence classes
- Salmon
- Estimating Posterior Uncertainty

Quantifying Uncertainty

Published online 2 August 2019

Nucleic Acids Research, 2019, Vol. 47, No. 18 e105

doi: 10.1093/nar/gkz622

Nonparametric expression analysis using inferential replicate counts

Anqi Zhu¹, Avi Srivastava², Joseph G. Ibrahim¹, Rob Patro² and Michael I. Love^{1,3,*}

¹Department of Biostatistics, University of North Carolina-Chapel Hill, 135 Dauer Drive, Chapel Hill, NC 27599, USA,

²Department of Computer Science, Stony Brook University, Computer Science Building, Engineering Dr, Stony Brook, NY 11794, USA and ³Department of Genetics, University of North Carolina-Chapel Hill, 120 Mason Farm Rd, Chapel Hill, NC 27514, USA

Inferential relative variance

$$\text{infRV}_{\mathbf{g}_i} = \frac{\max(0, \text{var}(\mathbf{g}_i) - \text{mean}(\mathbf{g}_i))}{\text{mean}(\mathbf{g}_i) + p} + s$$

Why (variance - mean) / mean?

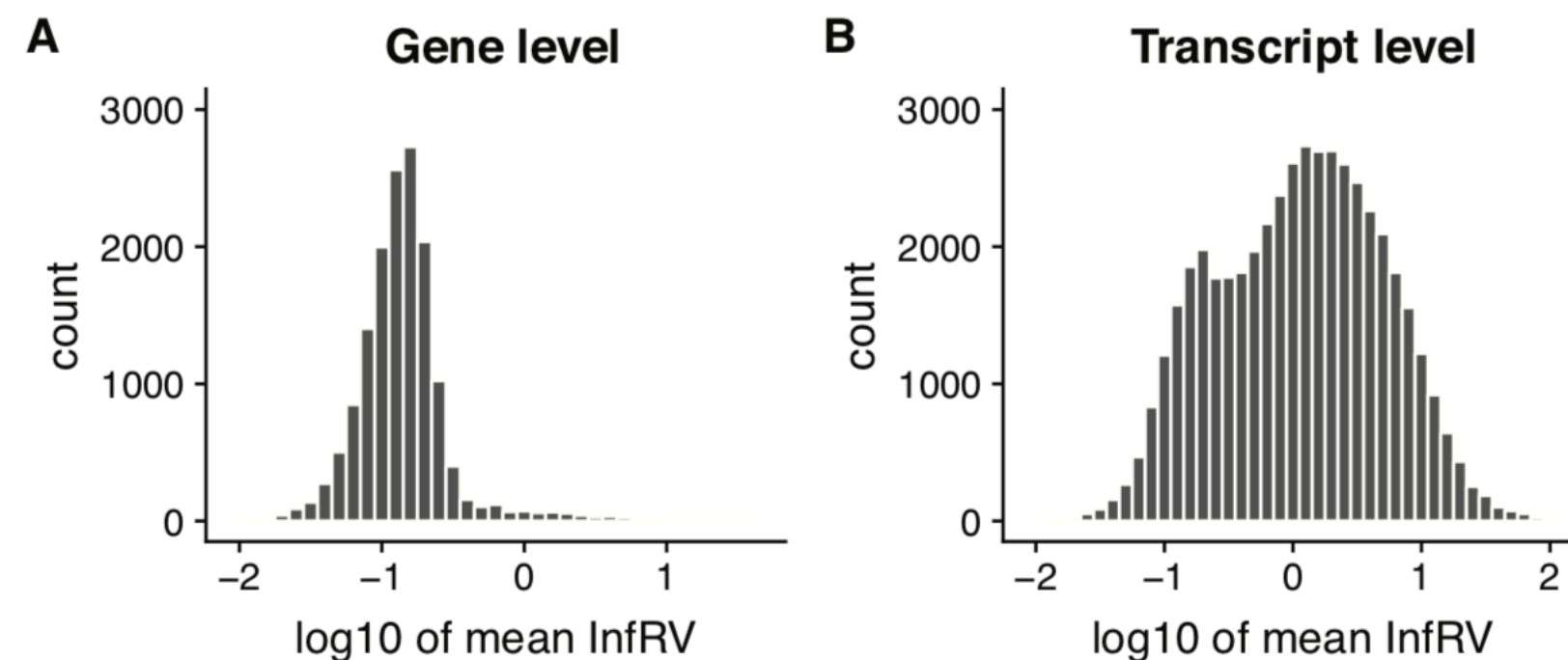
Minimally within a sample we have Poisson variance which scales with the mean.

So InfRV = ~0 with no multi-mapping.

Larger the value of InfRV, more uncertainty

Features for most downstream analysis

- People already perform analyses that greatly reduce such uncertainty; gene-level analysis
- But, gene-level analysis is often *too* coarse-grained.
- Many transcripts can be confidently quantified; some cannot; which transcripts are “impossible” differs from sample-to-sample.



We need “sub-gene” level transcriptional resolution

How should we, in general, address this problem?

What is the computational challenge here?

Let the data, themselves, tell you what the “inferentially distinguishable” transcriptional groups should be!

Using posterior uncertainty as a signal

As with MMSeq, Turro had an elegant solution to this as well!

Gene expression

Advance Access publication November 26, 2013

Flexible analysis of RNA-seq data using mixed effects models

Ernest Turro^{1,2,*}, William J. Astle³ and Simon Tavaré¹

¹Cancer Research UK Cambridge Institute, University of Cambridge, Robinson Way, Cambridge CB2 0RE, UK,

²Department of Haematology, University of Cambridge, NHS Blood and Transplant, Long Road, Cambridge CB2 0PT, UK and ³Department of Epidemiology, Biostatistics and Occupational Health, McGill University, 1020 Pine Avenue West, Montreal QC H3A 1A2, Canada

Associate Editor: Ivo Hofacker

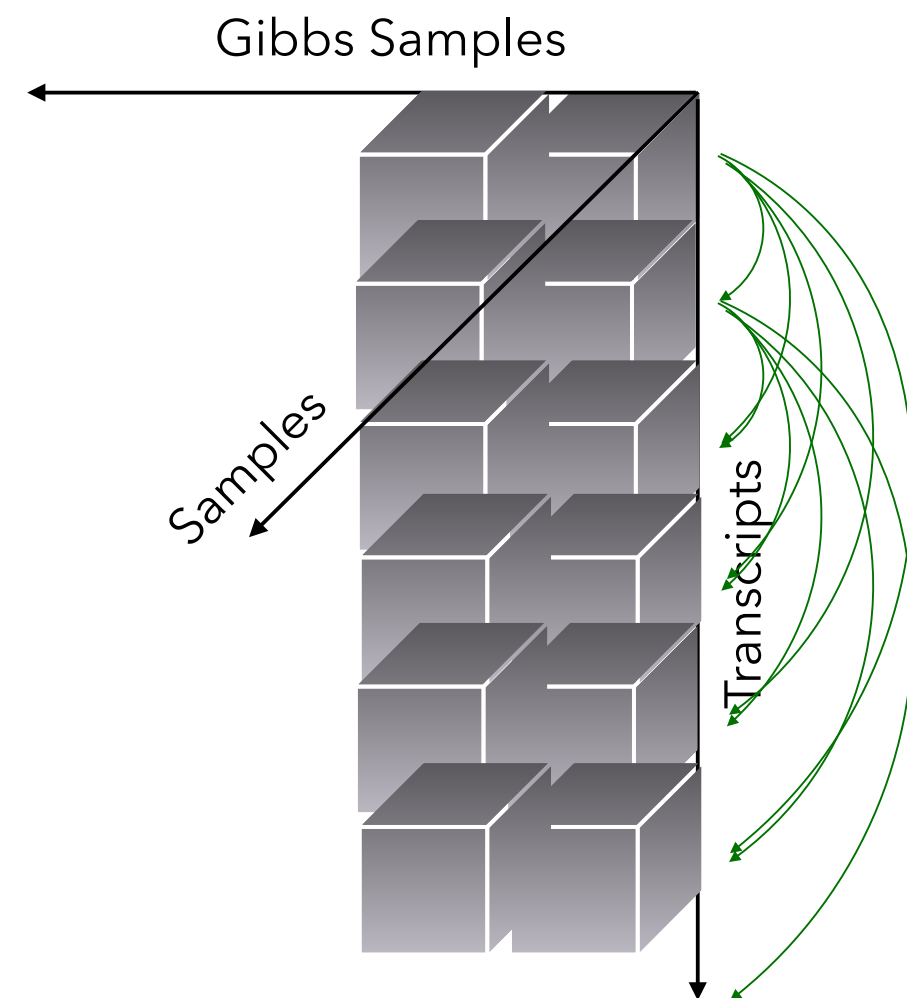
“ Often, sets of transcripts with poorly estimated expression parameters are anti-correlated because reads can be mapped to the combined set of transcripts with confidence, but no reads can be mapped to specific transcripts within the set. In these circumstances, it may be more informative to treat the set of transcripts as the unit of inference, rather than the individual transcripts.

We now propose an algorithm for collapsing transcripts with anti-correlated expression parameters into informative sets.

”

Using posterior uncertainty as a signal

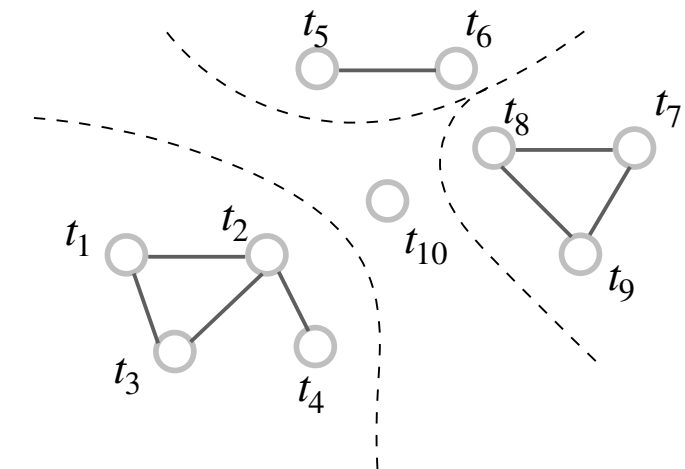
Unfortunately, this implementation tends to be impractical as transcriptome complexity increases, and also discovered some shortcomings with the particular approach.



mmcollapse

$$\binom{n}{2} \times m$$

This many correlation
calculation at the
first step



*Collapsing would
take place constrained
by the **fragment
ambiguity graph***

terminus

Terminus: Discovery of Data-driven Transcriptional Groups



Free & open-source tool (BSD3), available at :

<https://github.com/COMBINE-lab/Terminus>

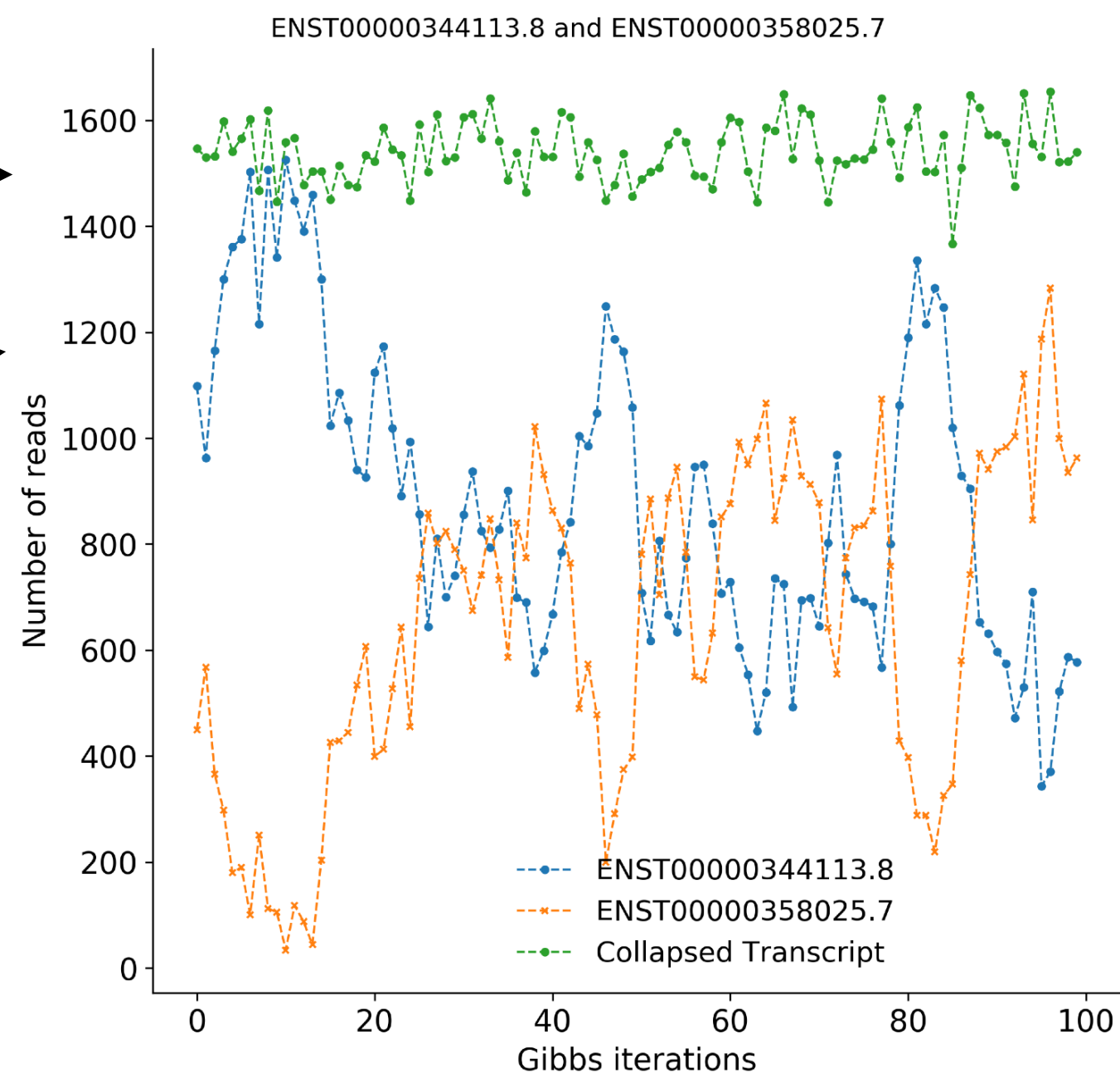


Defining transcriptional groups

Goal : Find groups of transcripts that, independently, have large inferential uncertainty, but, when grouped together, can be confidently quantified.

Actual Gibbs traces for a pair of transcripts merged by our algorithm.

Actual Gibbs traces for a pair of transcripts merged by our algorithm.

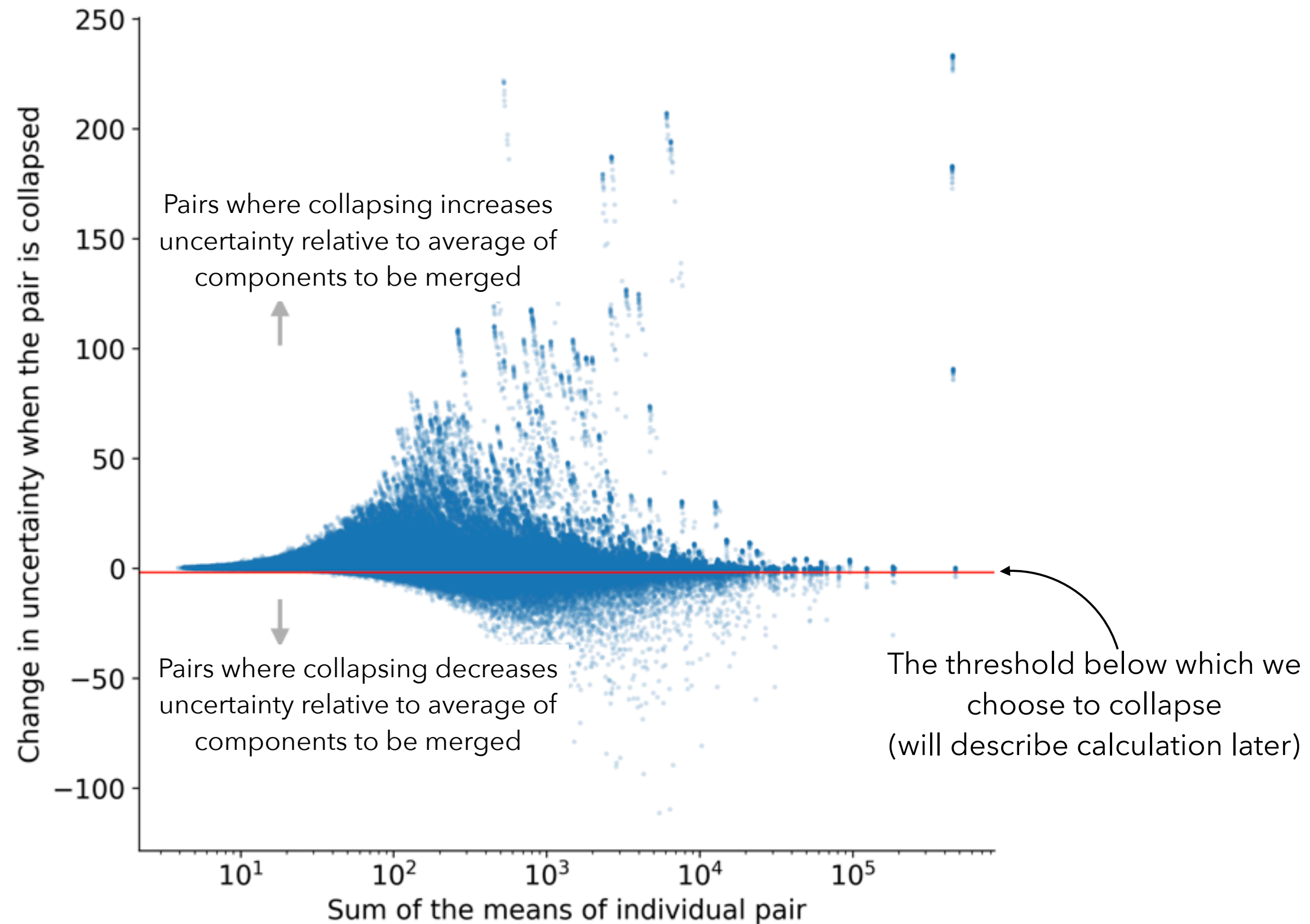


A metric for choosing transcripts to group

- We will employ an iterative algorithm that collapses groups that are “good candidates”; what makes a “good candidate”?
- We define the change of *InfRV* between the transcriptional group and average of individual group members as a metric to identify the promising transcripts to merge.
- For two transcripts (or previously defined groups) v_i and v_j we define the score for merging them as *InfRV* as :

$$s(v_i, v_j) = \text{infRV}(g_i + g_j) - \frac{\text{infRV}(g_i) + \text{infRV}(g_j)}{2}$$

How does this metric look?

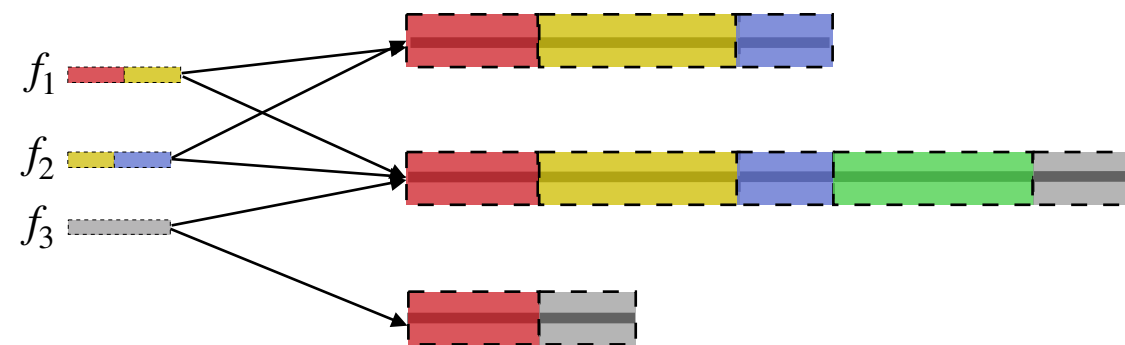


Evaluating all potential candidates is (far) too slow!

- For T transcripts there are $\binom{T}{2}$ pairs that should be considered.
- Not all pairs are important. In other words, almost all pairs should *not* be good candidates for collapse
- **Idea**: Use structure of alignment ambiguity to efficiently cull the search space!

Fragment Equivalence Classes and the Fragment Ambiguity Graphs

Mapping Information



A
B
C

Transcripts

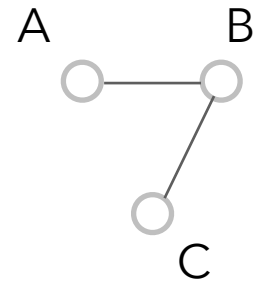
Equivalence Relation

$f_1 \rightarrow (A, B)$
 $f_2 \rightarrow (A, B)$
 $f_3 \rightarrow (B, C)$

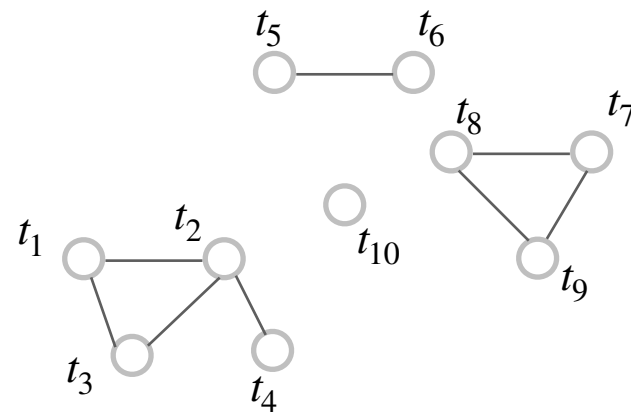
Equivalence Class

$(A, B) \rightarrow 2$
 $(B, C) \rightarrow 1$

Network



The resulting network (graph) describes the “fragment-level” ambiguity



$\mathcal{E}_j : \{t_5, t_6\} \rightarrow \{w_{5,j}, w_{6,j}; c\}$

Note: An edge can be labeled with > 1 equivalence class.

Rich Equivalence Class Attributes

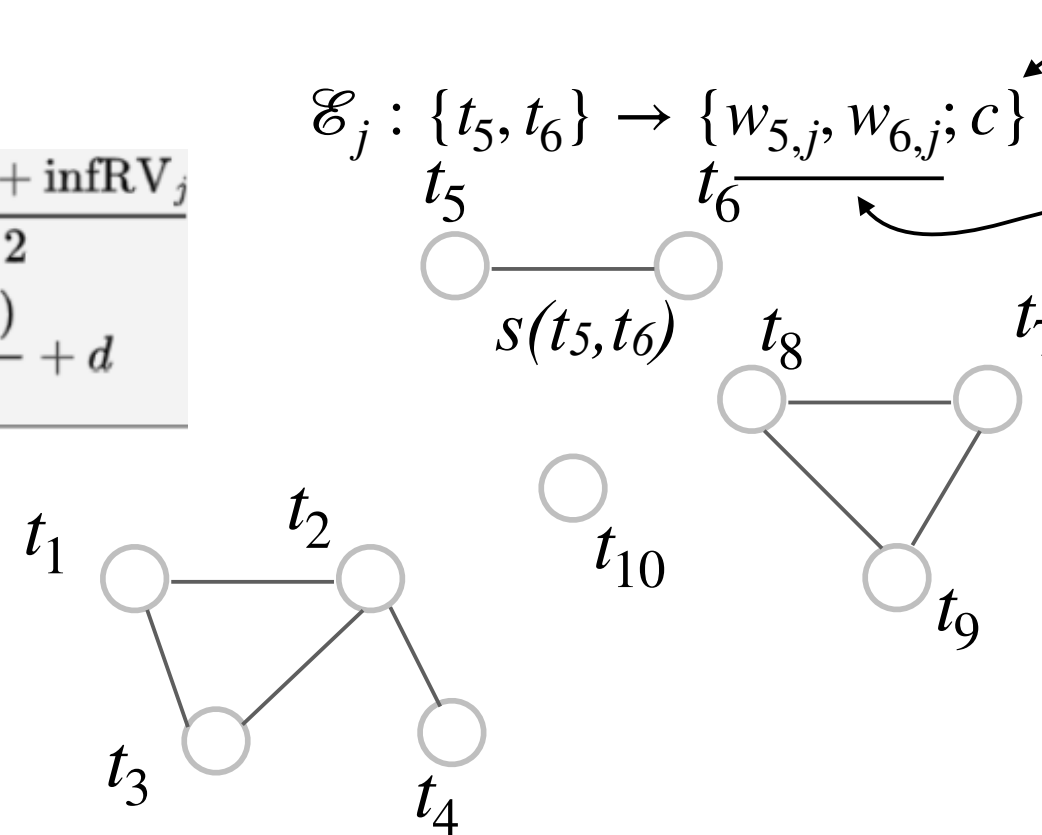
- Equivalence Class Count
- Conditional probabilities $w_{i,j}$ for transcript i in equivalence class j . It consolidates the probability of $\sum_{f_k \in \mathcal{E}_j} P\{f_k | t_i\}$

Fragment Equivalence Classes and the Fragment Ambiguity Graphs

$$s(t_i, t_j) = \text{infRV}_{i+j} - \frac{\text{infRV}_i + \text{infRV}_j}{2}$$

$$\text{infRV}_i = \frac{\max(\sigma_{pi}^2 - \mu_i, 0)}{\mu_i + pc} + d$$

The lower the edge weight, the better are those candidates for grouping.



- Equivalence Class Count
- Conditional probabilities $w_{i,j}$ for transcript i in equivalence class j . It consolidates the probability of $\sum_{f_k \in \mathcal{E}_j} P\{f_k | t_i\}$

Construct a graph $G = (V, E)$ where :

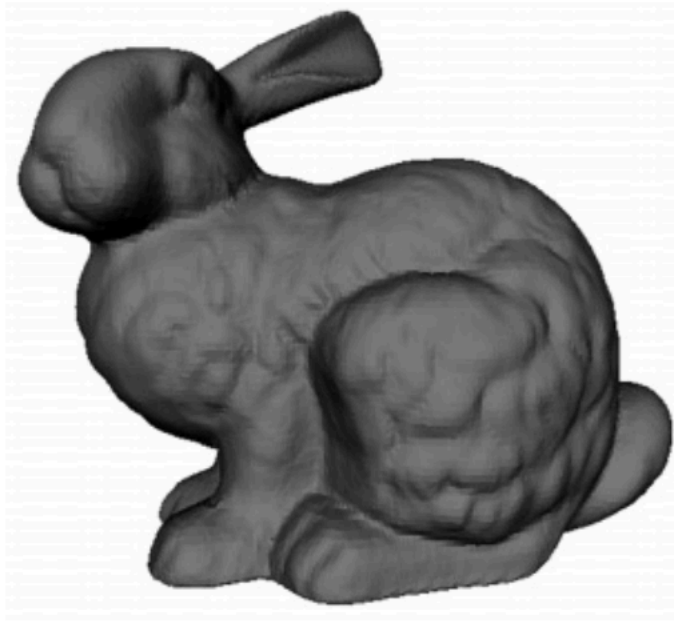
V = set of transcripts

$E = \{v_i, v_j\}$ such that v_i, v_j co-occur within at least one equivalence class

Motivation: Transcripts sharing *no* reads are unlikely to be good candidates for merging

Discovering transcriptional groups is equivalent to contracting edges in this graph!

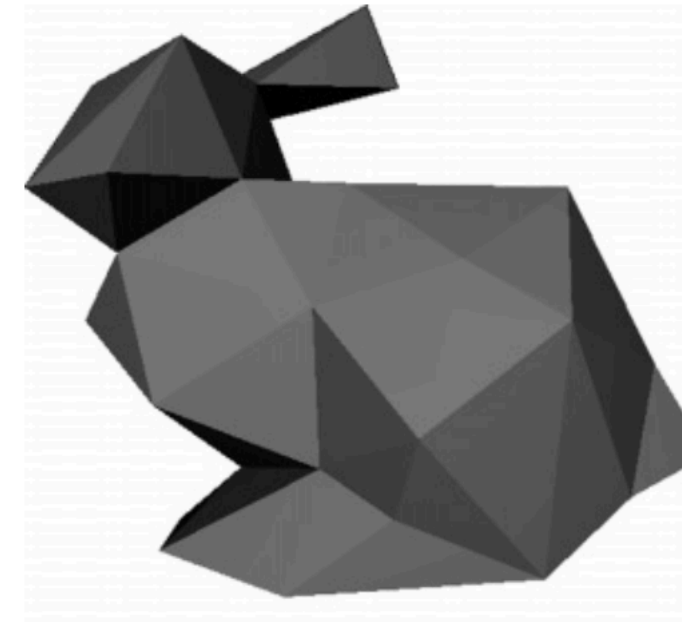
Some interdisciplinary motivation



69,451 triangles to render
original bunny

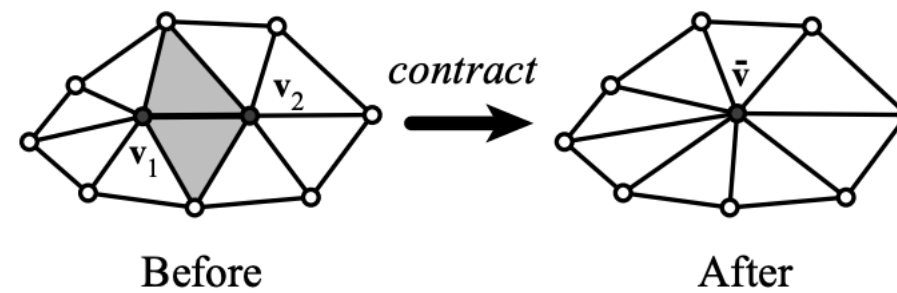


Approximation with 1000
triangles

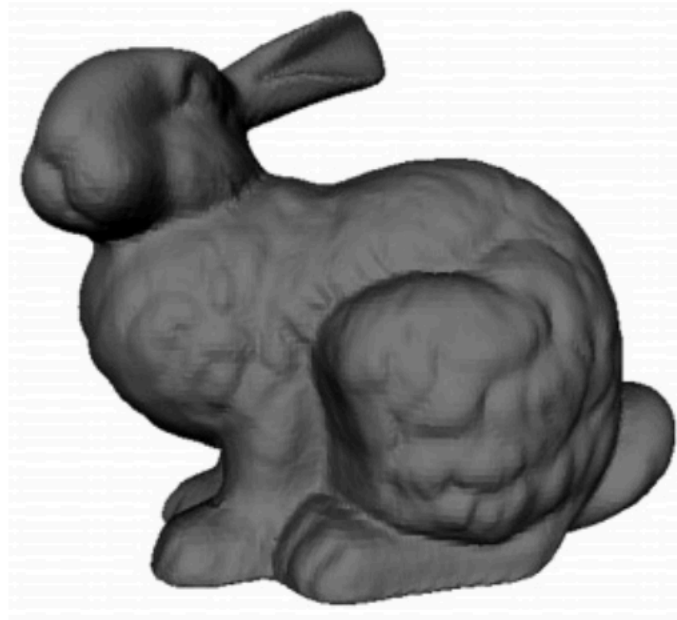


Approximation with 100
triangles

Goal : Simplify mesh by reducing # of vertices while maintaining visual fidelity.



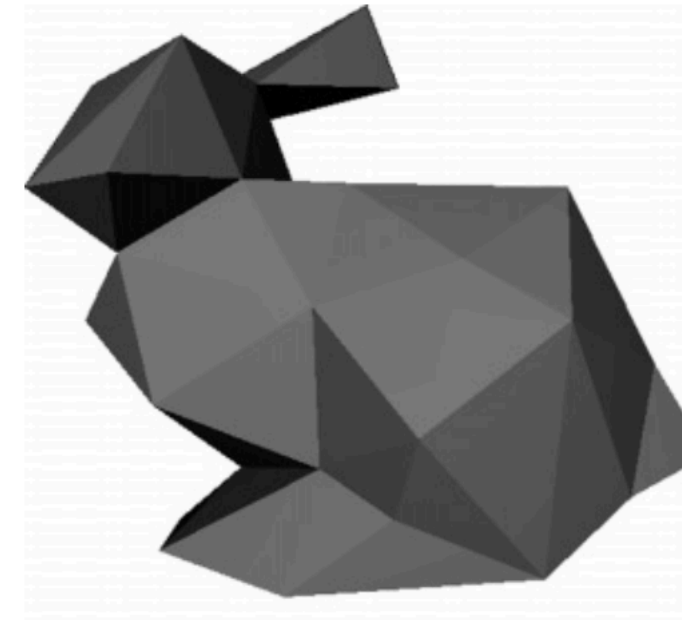
Some interdisciplinary motivation



69,451 triangles to render
original bunny



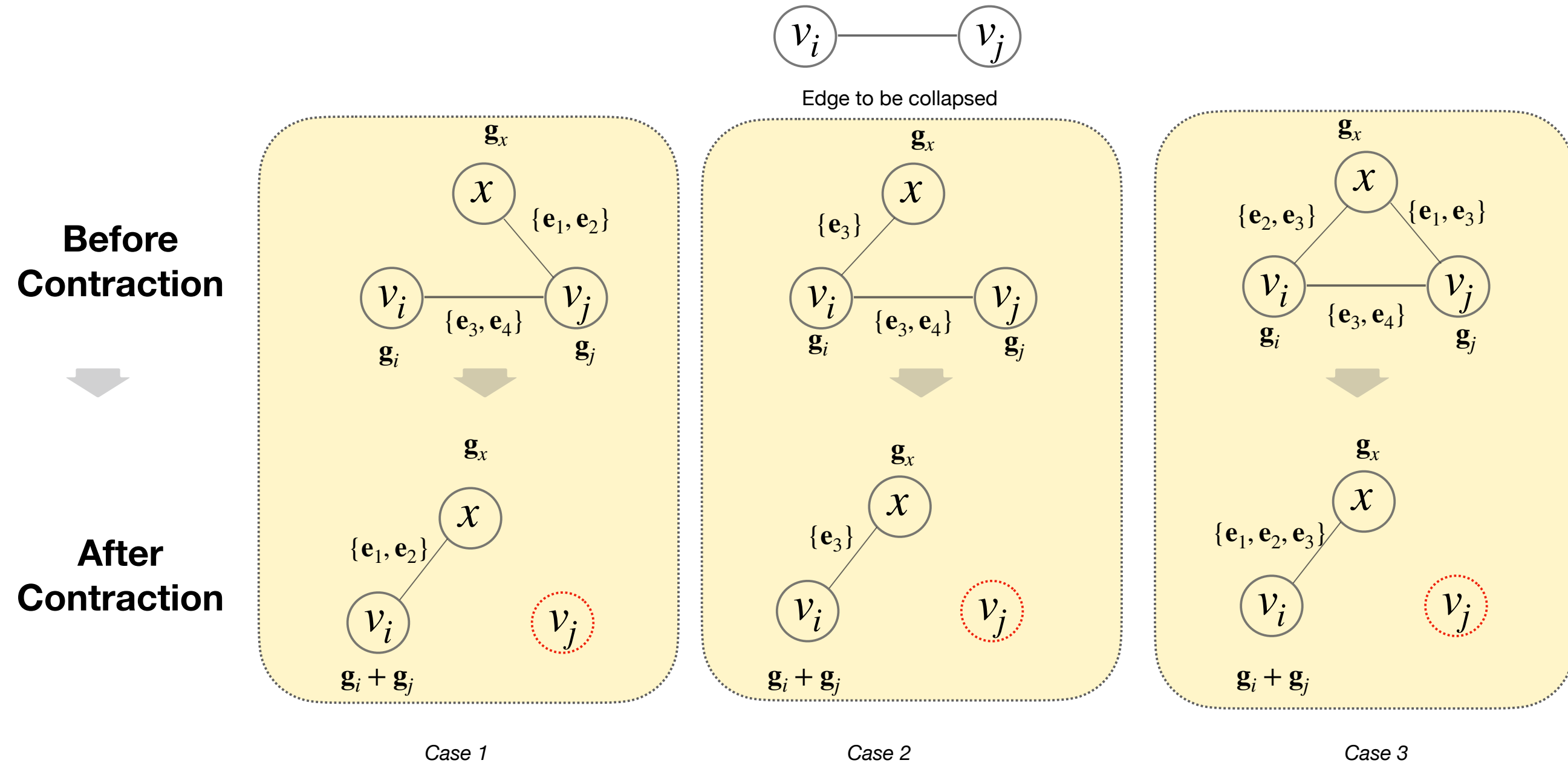
Approximation with 1000
triangles



Approximation with 100
triangles

Approach : Put all edges in a [min heap](#). While we have not exceeded our approximation budget, extract the edge whose collapse will make the *least* difference, contract the edge, evaluate the newly adjacent vertices, and place them back in the heap.

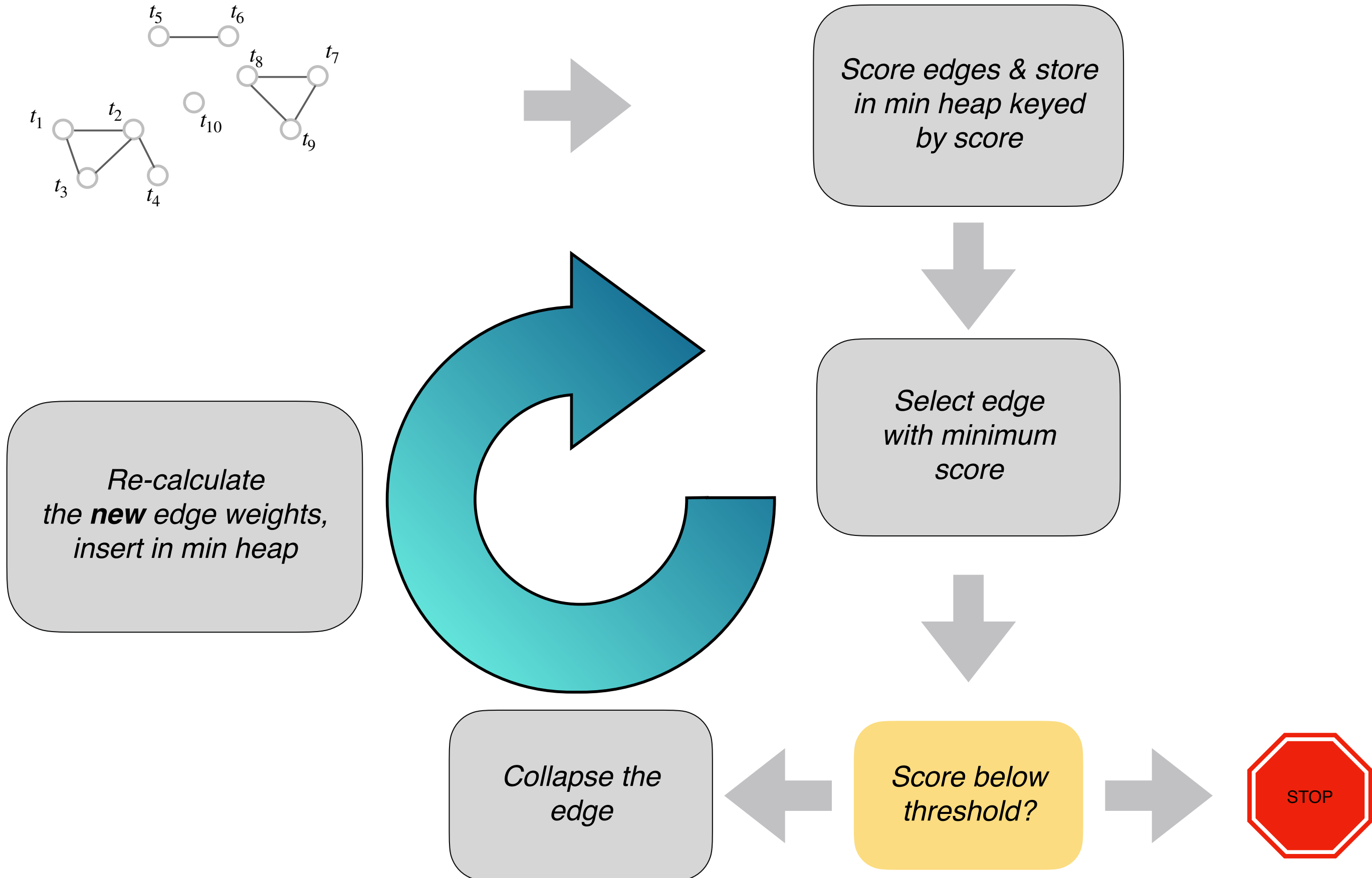
How to "contract" an edge



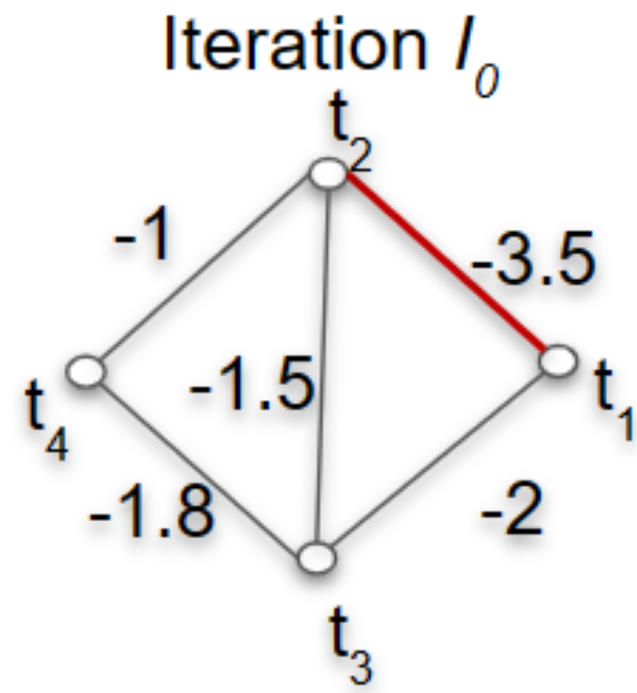
Updated Score

$$s(v_i, x) = \text{infRV}(\mathbf{g}_i + \mathbf{g}_j + \mathbf{g}_x) - \frac{\text{infRV}(\mathbf{g}_i + \mathbf{g}_j) + \text{infRV}(\mathbf{g}_x)}{2}$$

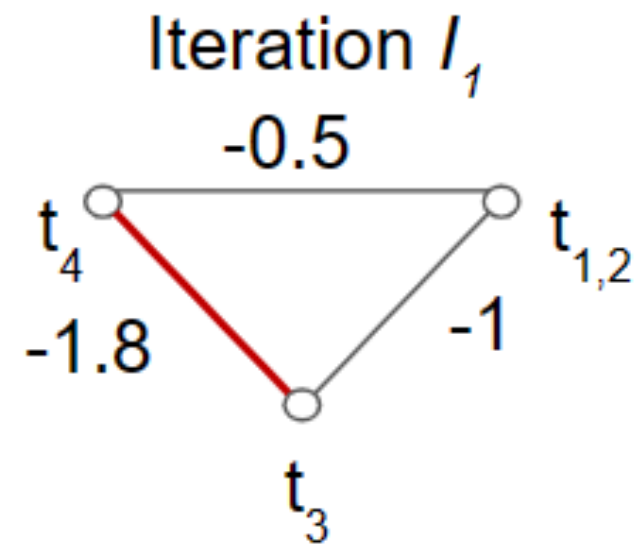
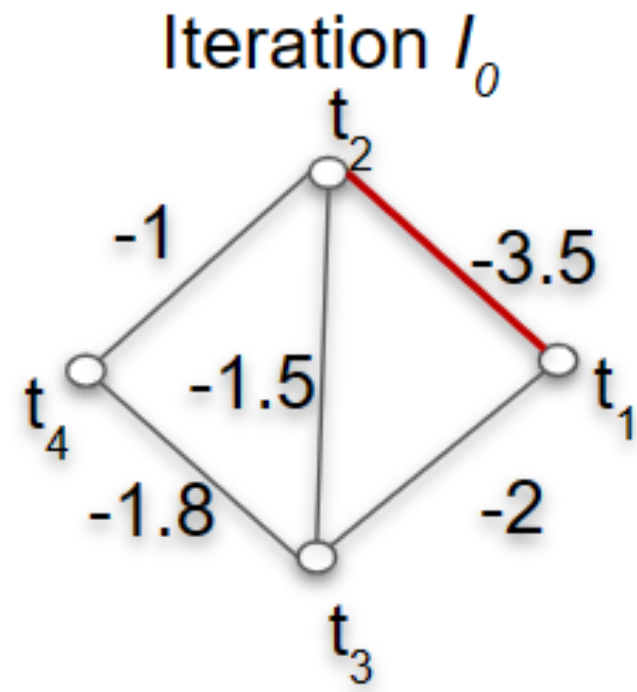
Overview of the algorithm



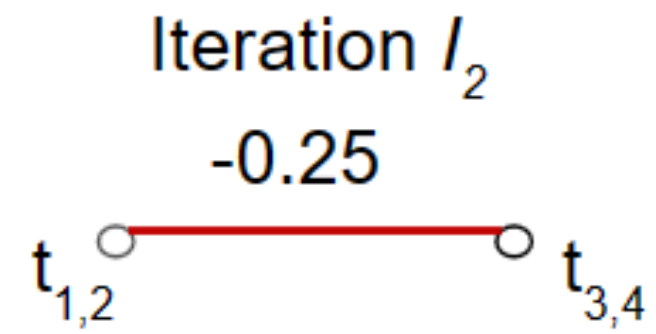
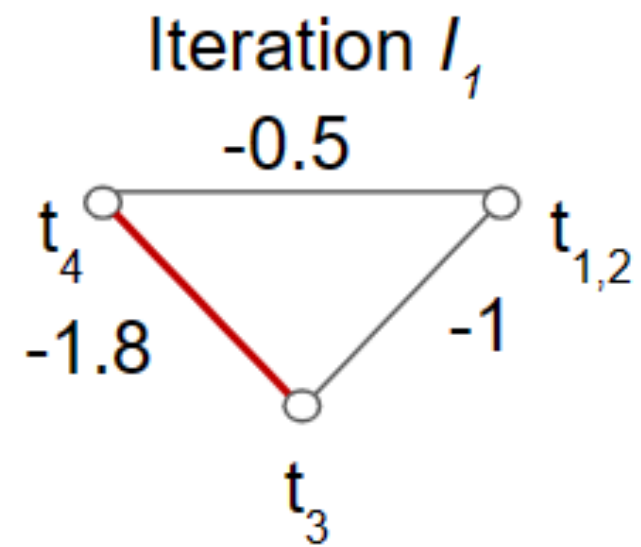
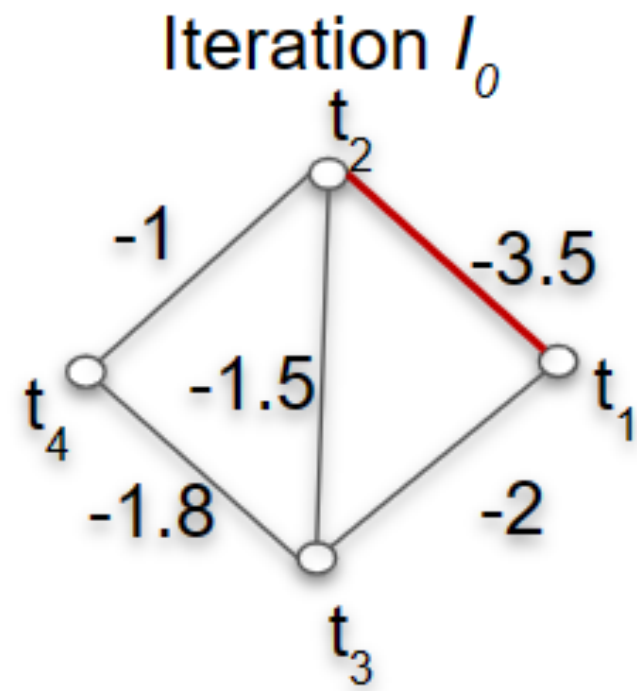
Example



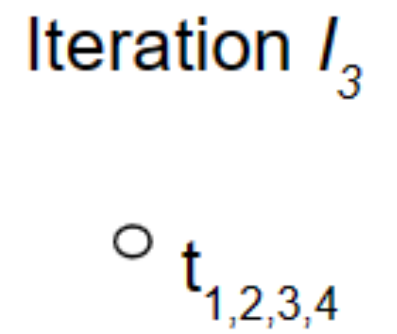
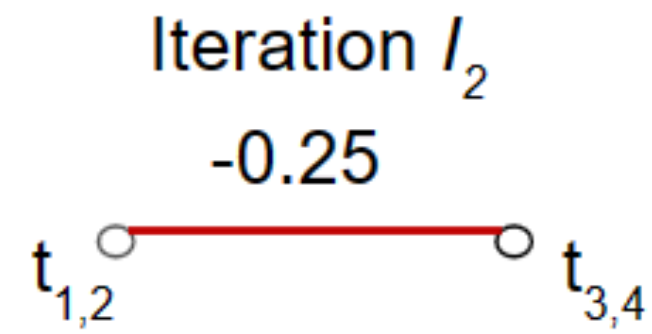
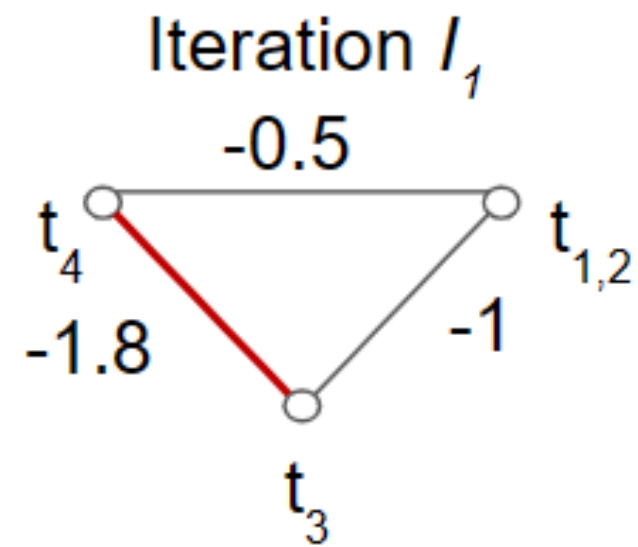
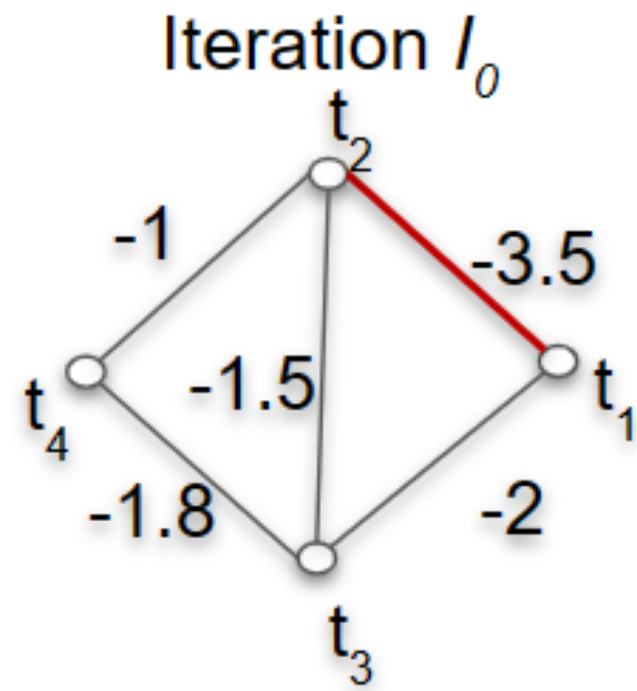
Example



Example



Example



Determining when to stop!

Data: $\mathcal{T}, \mathcal{G}$, selection_size

Result: τ_{new}

$\tau_{old} \leftarrow 0$;

$\tau_{new} \leftarrow 0$;

converged $\leftarrow$ False ;

$S \leftarrow \{(i,j) | i,j \in \mathcal{T}\}, s.t. |S| = \text{selection_size}$;

$\mathcal{B} \leftarrow \emptyset$;

while not converged **do**

for $(i,j) \in S$ **do**

$\mathcal{B} = \mathcal{B} \cup s(i,j)$;

end

$\tau_{new} \leftarrow \text{percentile}(\mathcal{B}, 2.5)$;

 diff $\leftarrow |\tau_{old} - \tau_{new}|$;

if diff $< \delta$ **then**

 converged $\leftarrow$ True ;

end

 selection_size $\leftarrow$ selection_size $\cdot 2$;

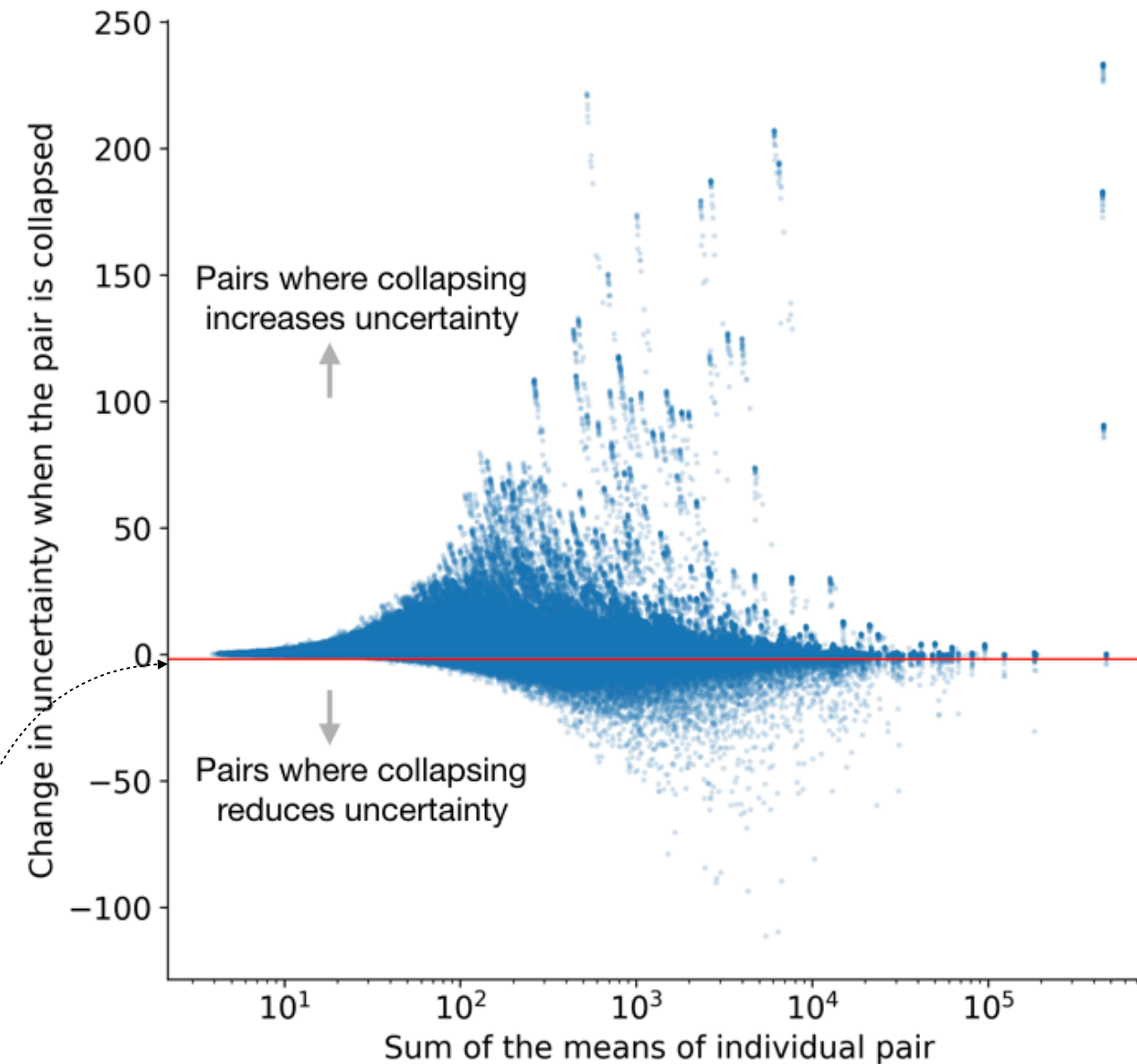
$\tau_{old} \leftarrow \tau_{new}$;

$\mathcal{B} \leftarrow \emptyset$;

end

return τ_{new} ;

Algorithm 1: Threshold τ selection algorithm



Determining when to stop!

Data: $\mathcal{T}, \mathcal{G}$, selection_size

Result: τ_{new}

$\tau_{old} \leftarrow 0$;

$\tau_{new} \leftarrow 0$;

converged $\leftarrow$ False ;

$S \leftarrow \{(i,j) | i,j \in \mathcal{T}\}, s.t. |S| = \text{selection_size}$;

$B \leftarrow \emptyset$;

while not converged do

for $(i,j) \in S$ do

$B = B \cup s(i,j)$;

end

$\tau_{new} \leftarrow \text{percentile}(B, 2.5)$;

diff $\leftarrow |\tau_{old} - \tau_{new}|$;

if diff $< \delta$ then

 converged $\leftarrow$ True ;

end

selection_size $\leftarrow$ selection_size $\cdot 2$;

$\tau_{old} \leftarrow \tau_{new}$;

$B \leftarrow \emptyset$;

end

return τ_{new} ;

Algorithm 1: Threshold τ selection algorithm

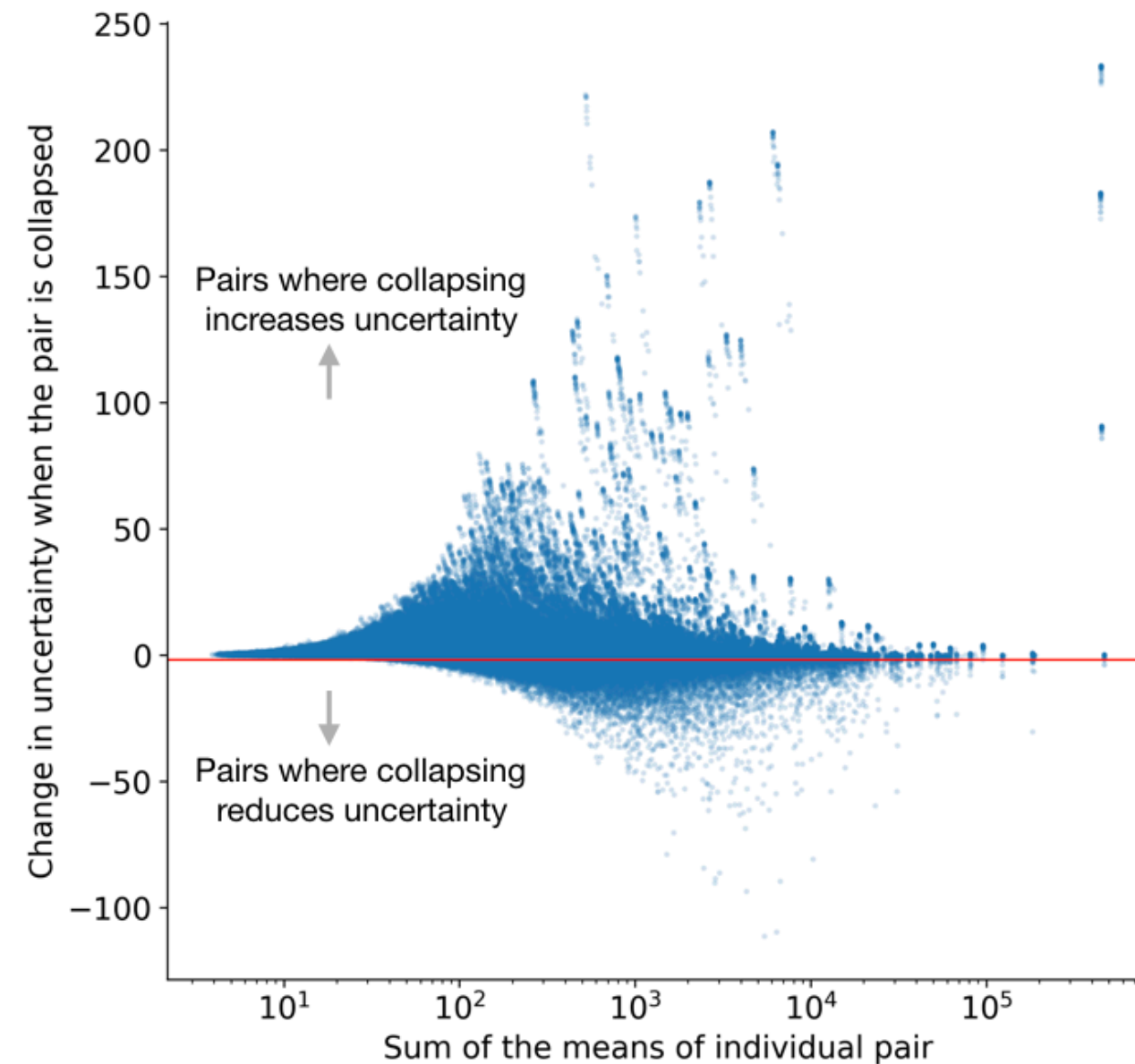
Basically:

Define the background distribution as the distribution of “scores” for random transcript pairs.

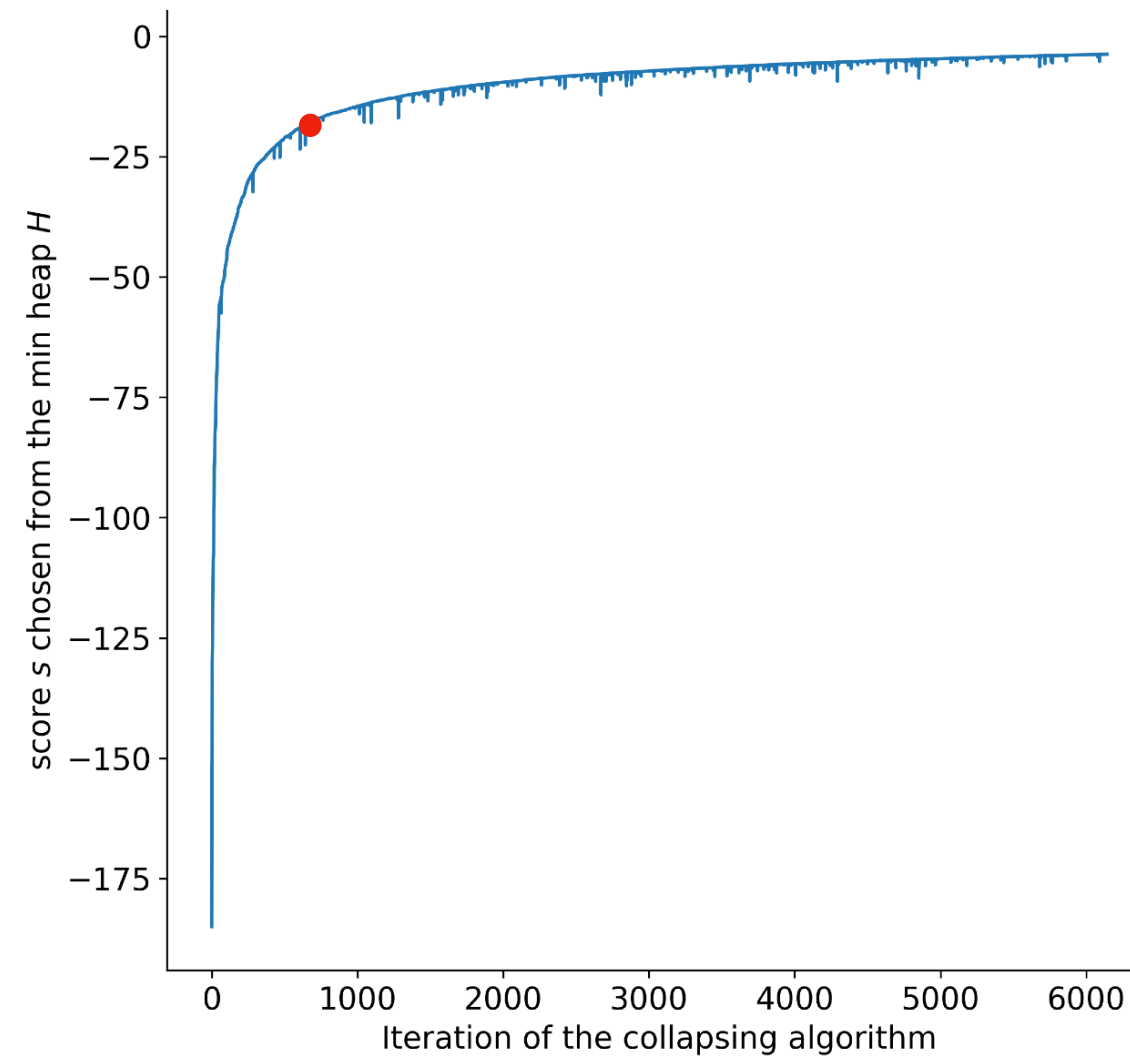
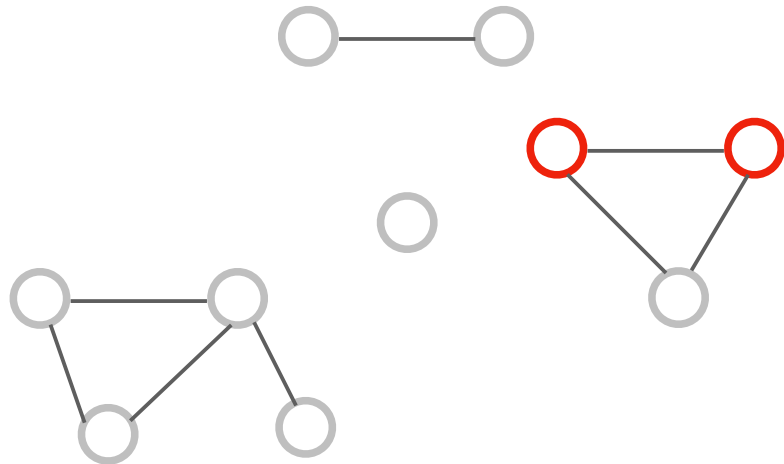
Estimate the tail (2.5%) of this distribution.

Repeatedly double the size of the background distribution until this quantile stops changing.

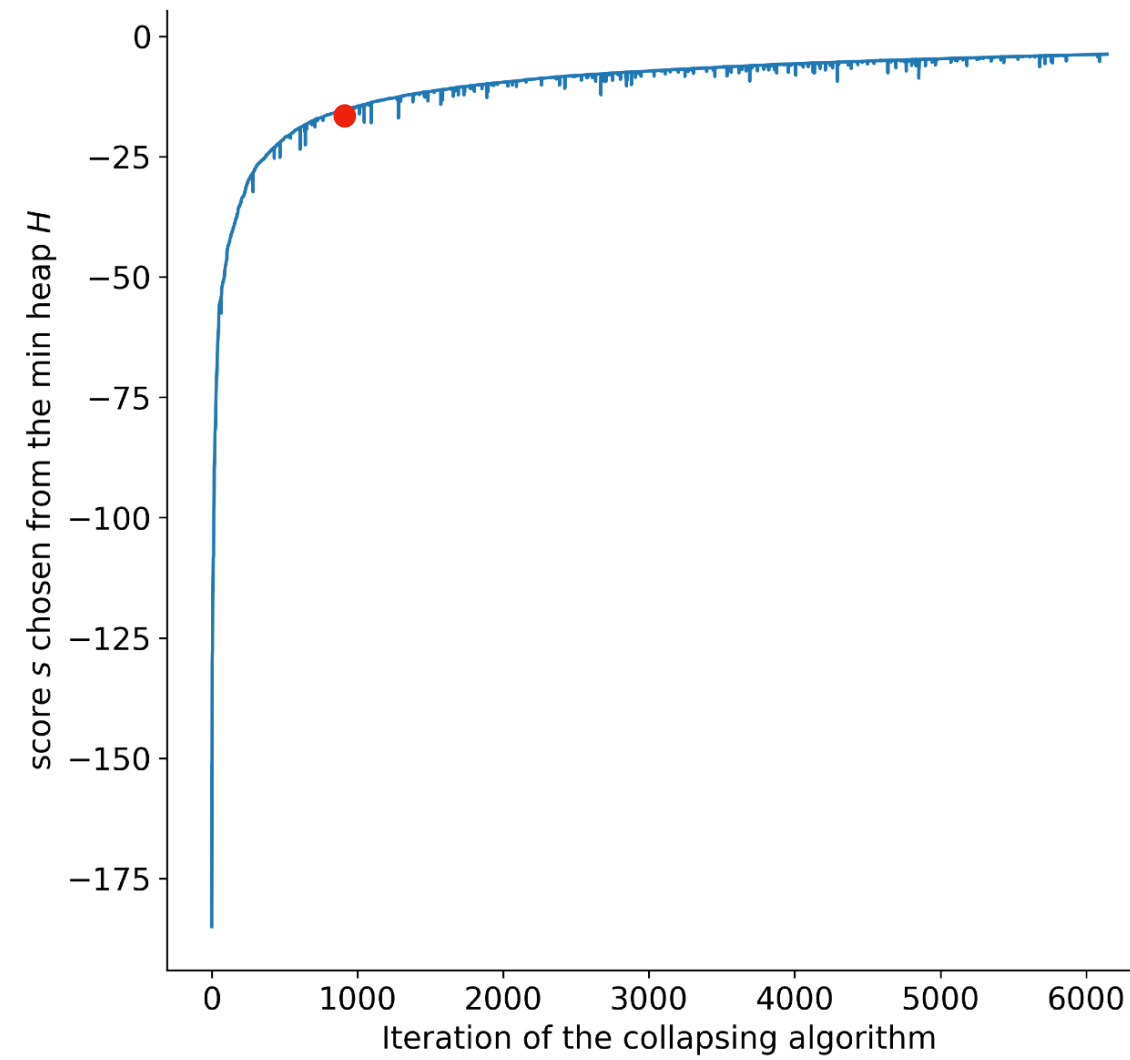
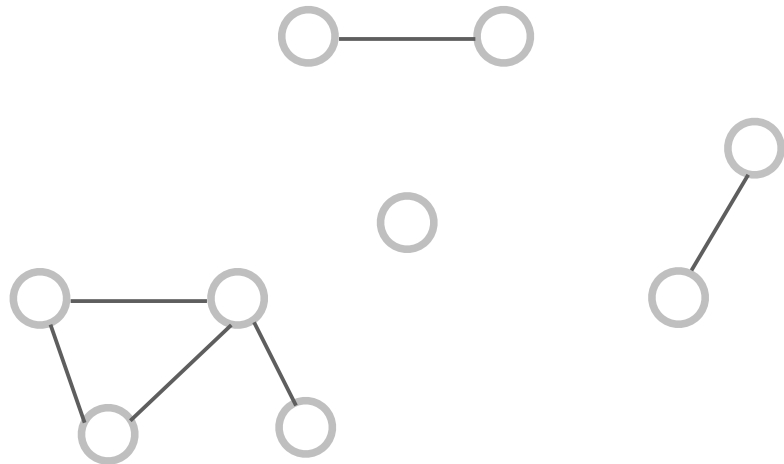
This is our threshold.



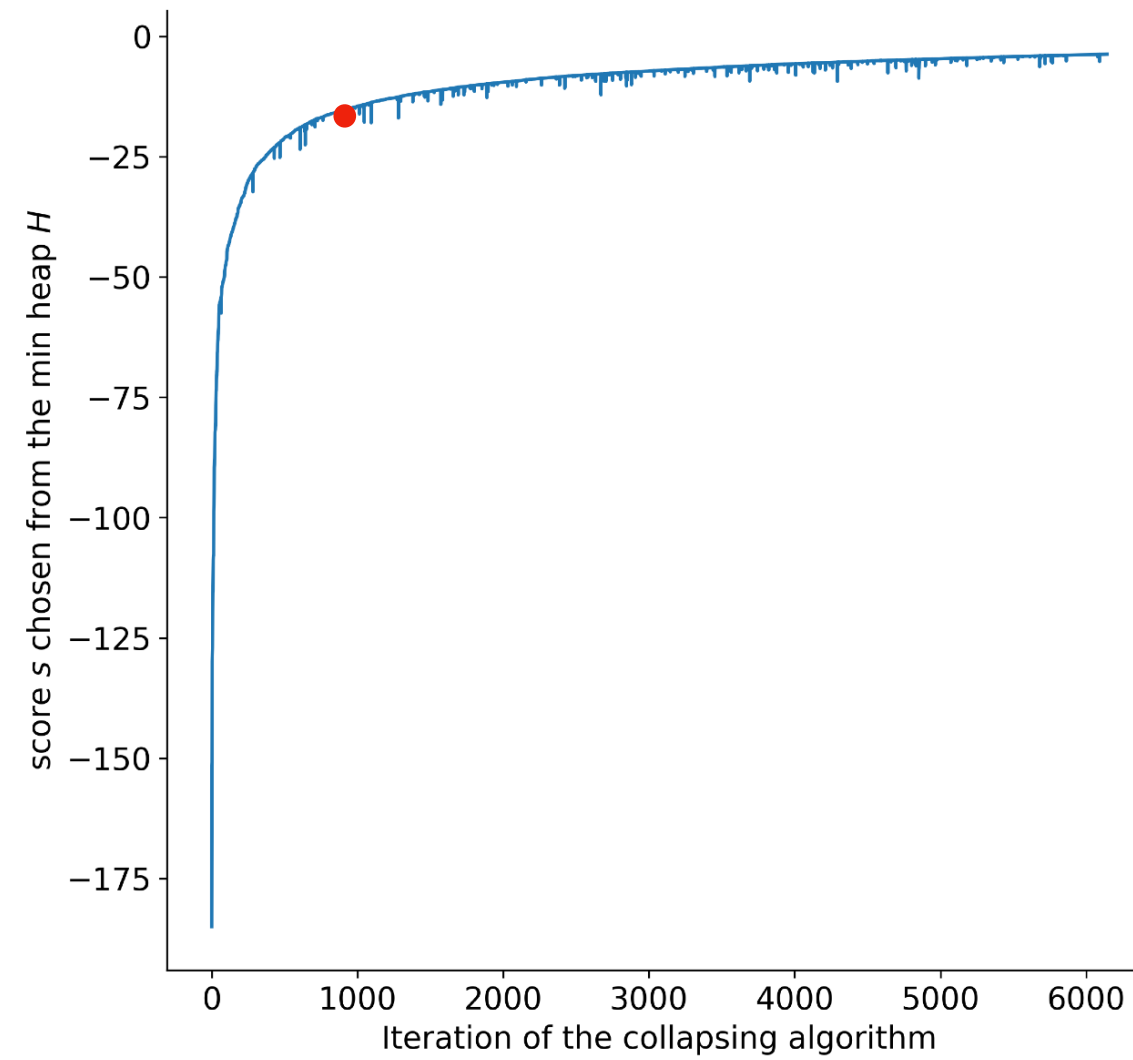
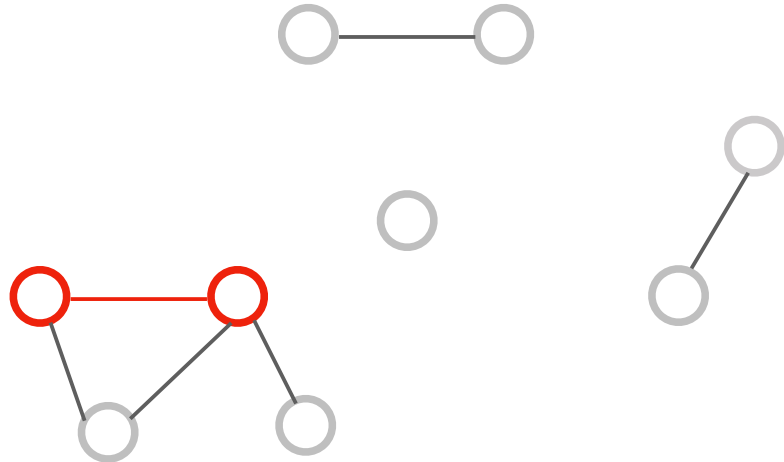
Progress over iterations



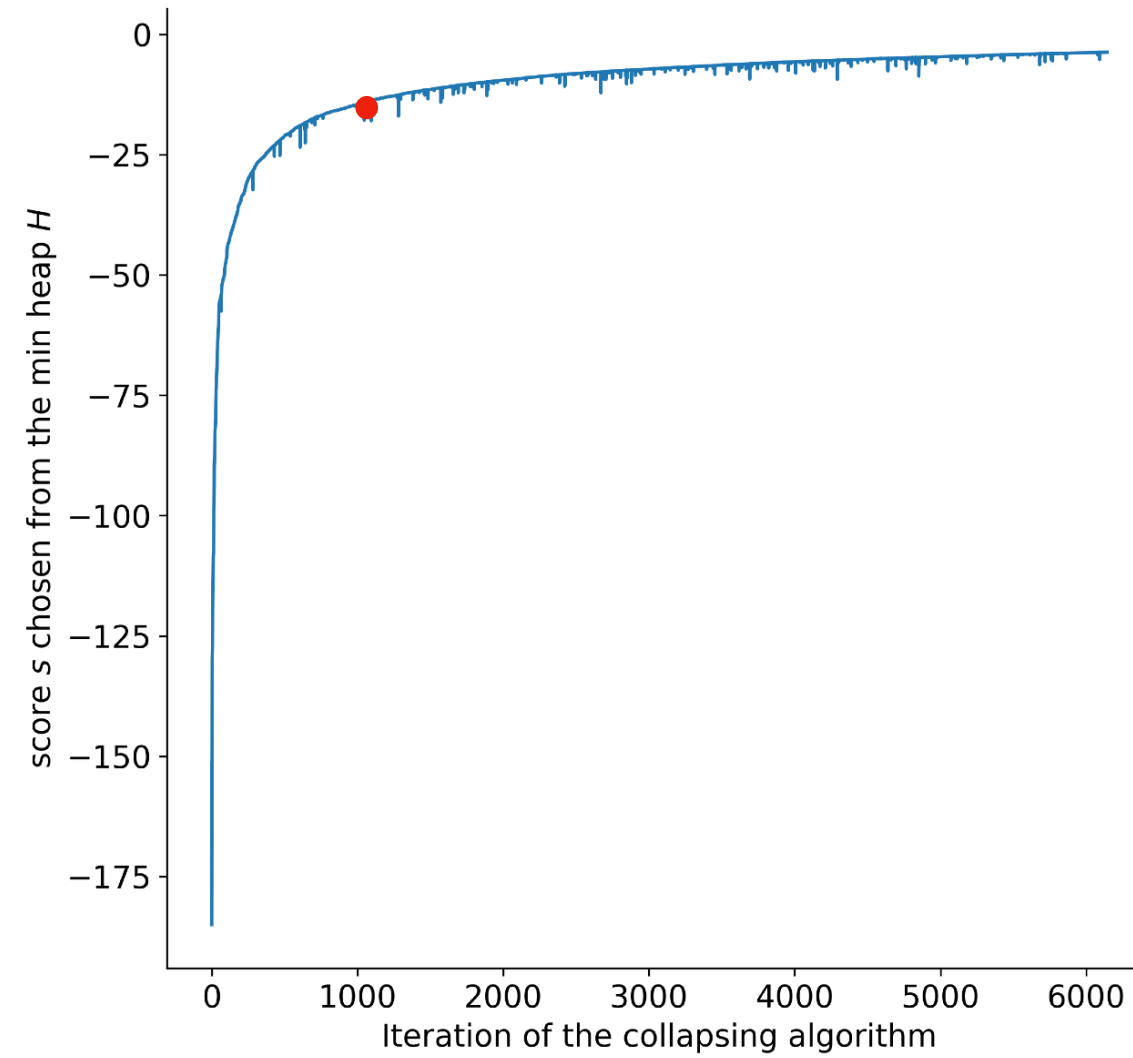
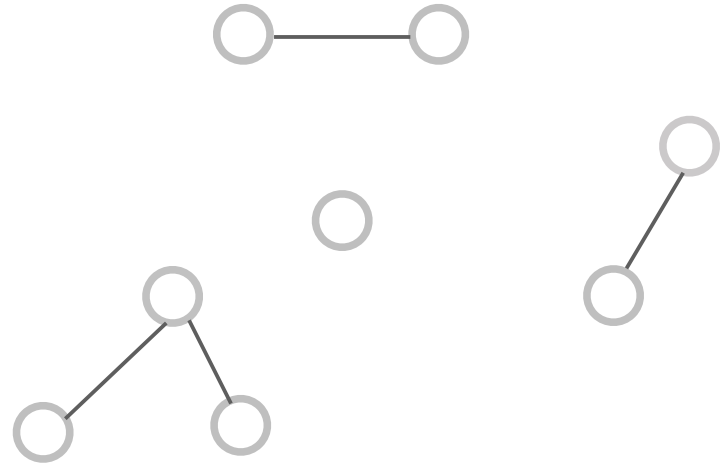
Progress over iterations



Progress over iterations



Progress over iterations



Terminus for multiple samples

So far the algorithm we have presented, works for only a single RNA-Seq sample. This step is called for a group step. The output is a set of discrete transcript groups.

However, an RNA-Seq experiment consists of multiple samples. We want such discrete transcript groups for an experiment across multiple samples. Terminus provides a **consensus** step.

Consensus

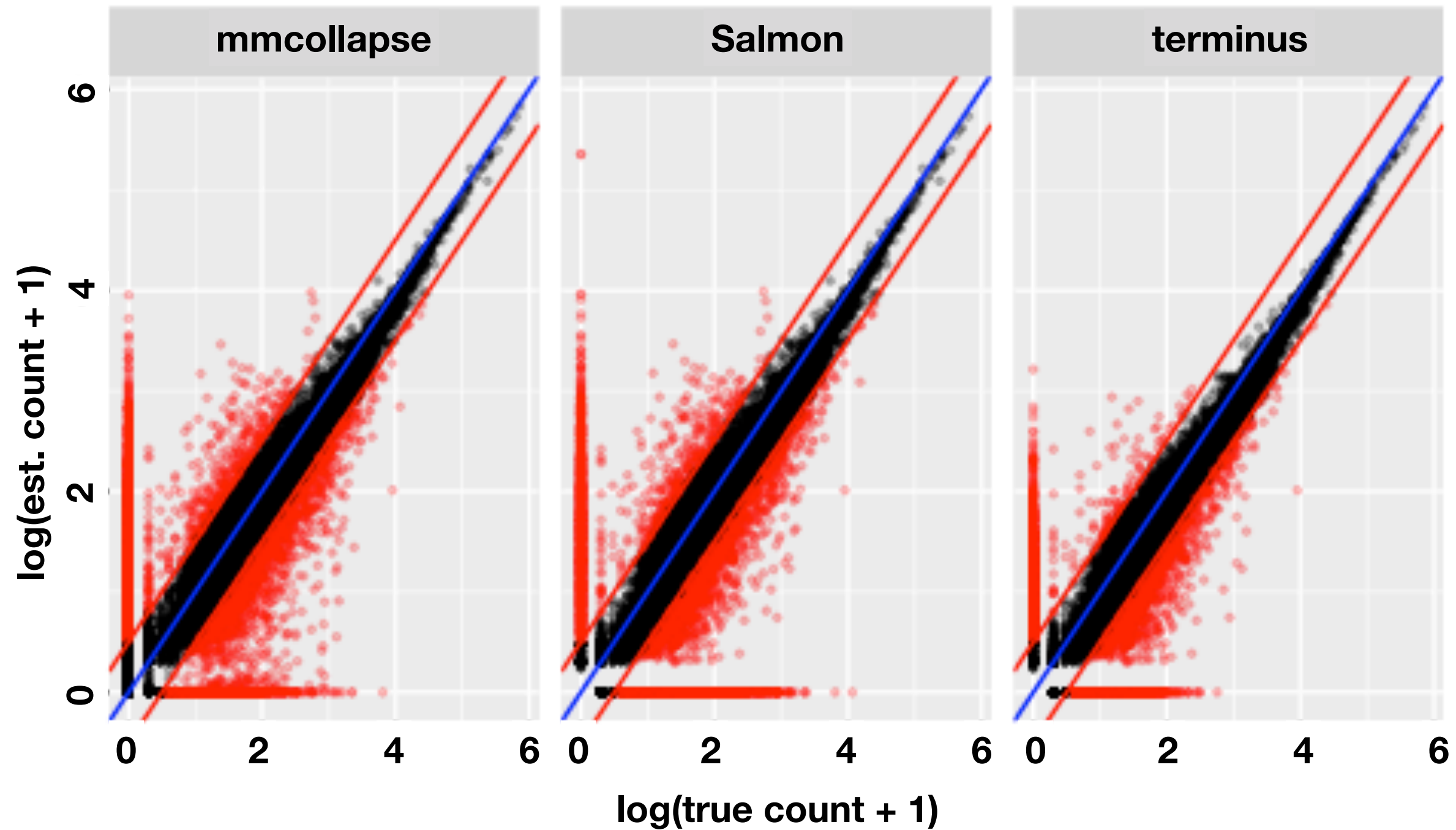
The consensus procedure starts with individual groups, and constructs a union graph by treating each of the groups as a complete connected undirected graph (clique). For example: given two different groups $\{v_i, v_j, v_k\}$ from one sample and $\{v_i, v_j\}$ from another, terminus first constructs a weighted triangle with end points v_i, v_j and v_k where each edge has a weight 1. While considering the second group, terminus increases the weight of edge $\{v_i, v_j\}$ by 1. This iterative procedure generates a weighted graph ensuring any edge in the graph represents two transcripts that belong to the same group in at least one sample. Terminus further prunes the union graph and removes the edges that have a weight below a user-defined threshold. The consensus mechanism ensures that a pair of transcripts should at least co-occur in a specified number of samples to qualify for the final grouping.

How do terminus groups compare to individual transcripts?

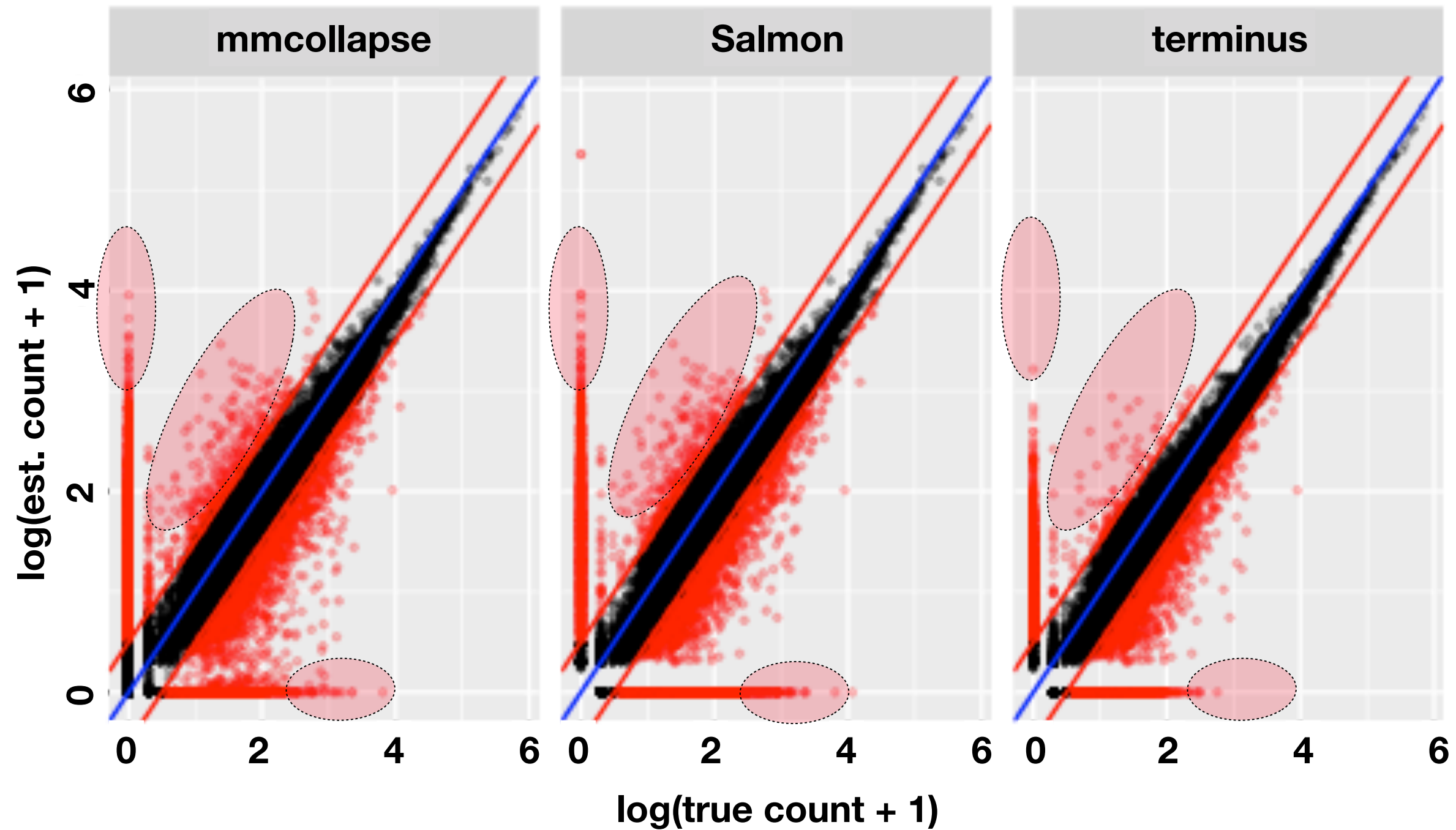
To evaluate we looked at simulated & experimental data

- full-transcriptome simulated data in Human
multi-sample 4x4
- experimental data in Drosophila
subset of "pasilla"* dataset

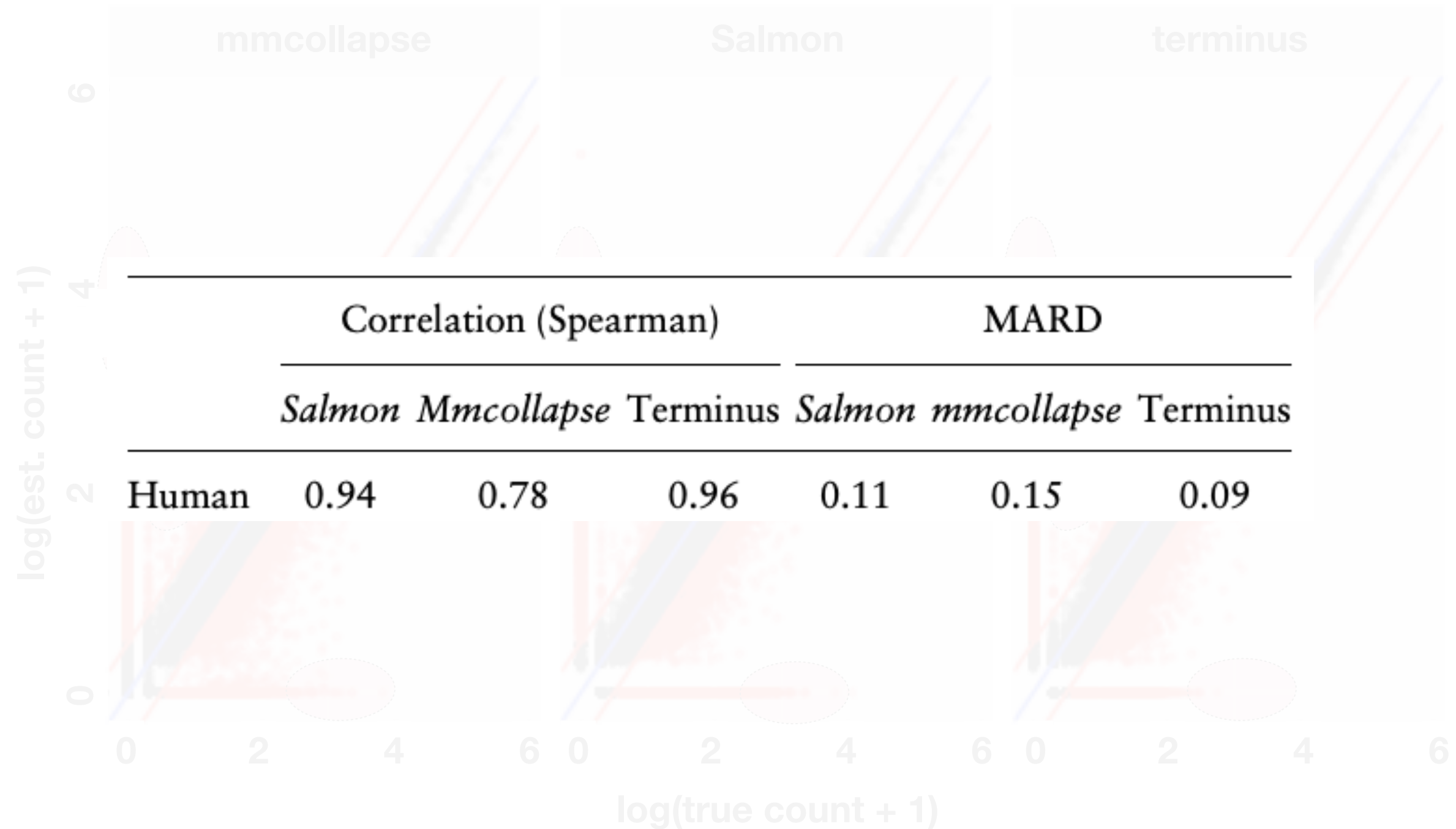
How do terminus groups compare to individual transcripts?



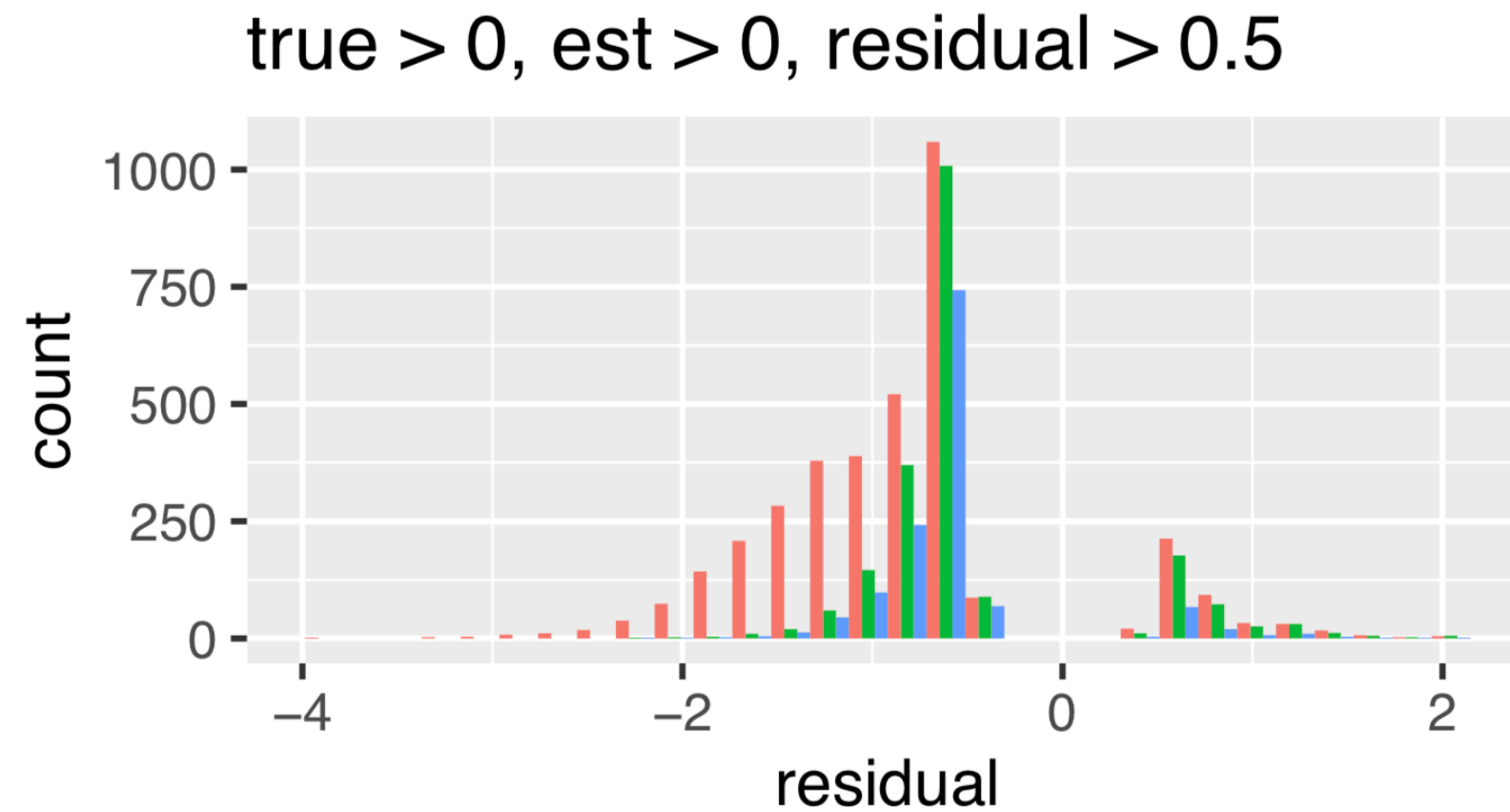
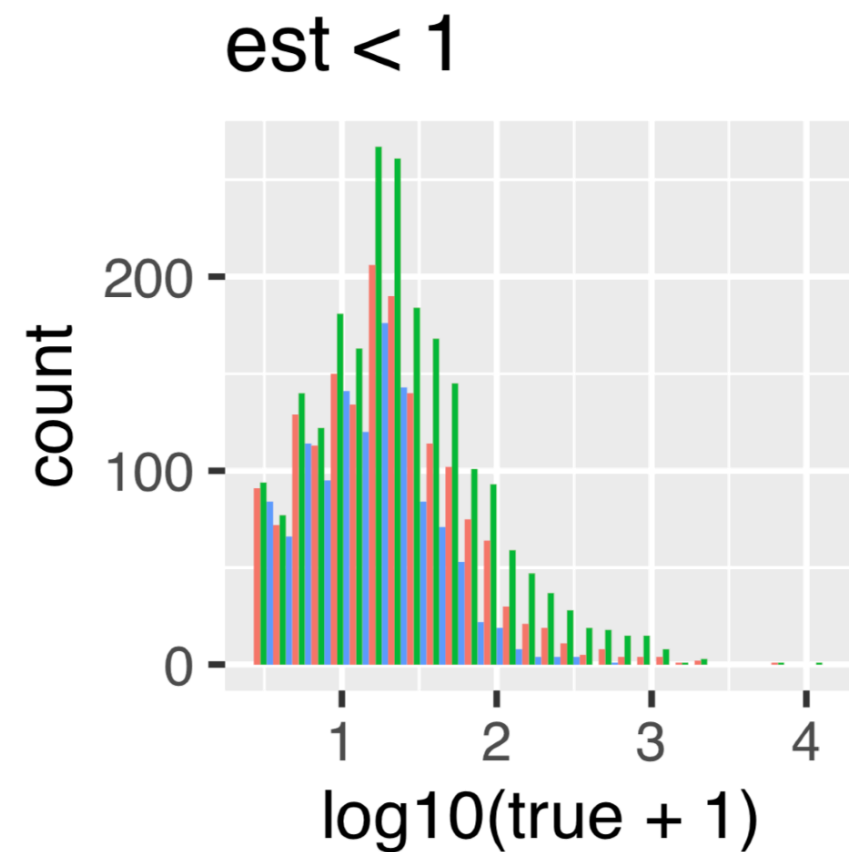
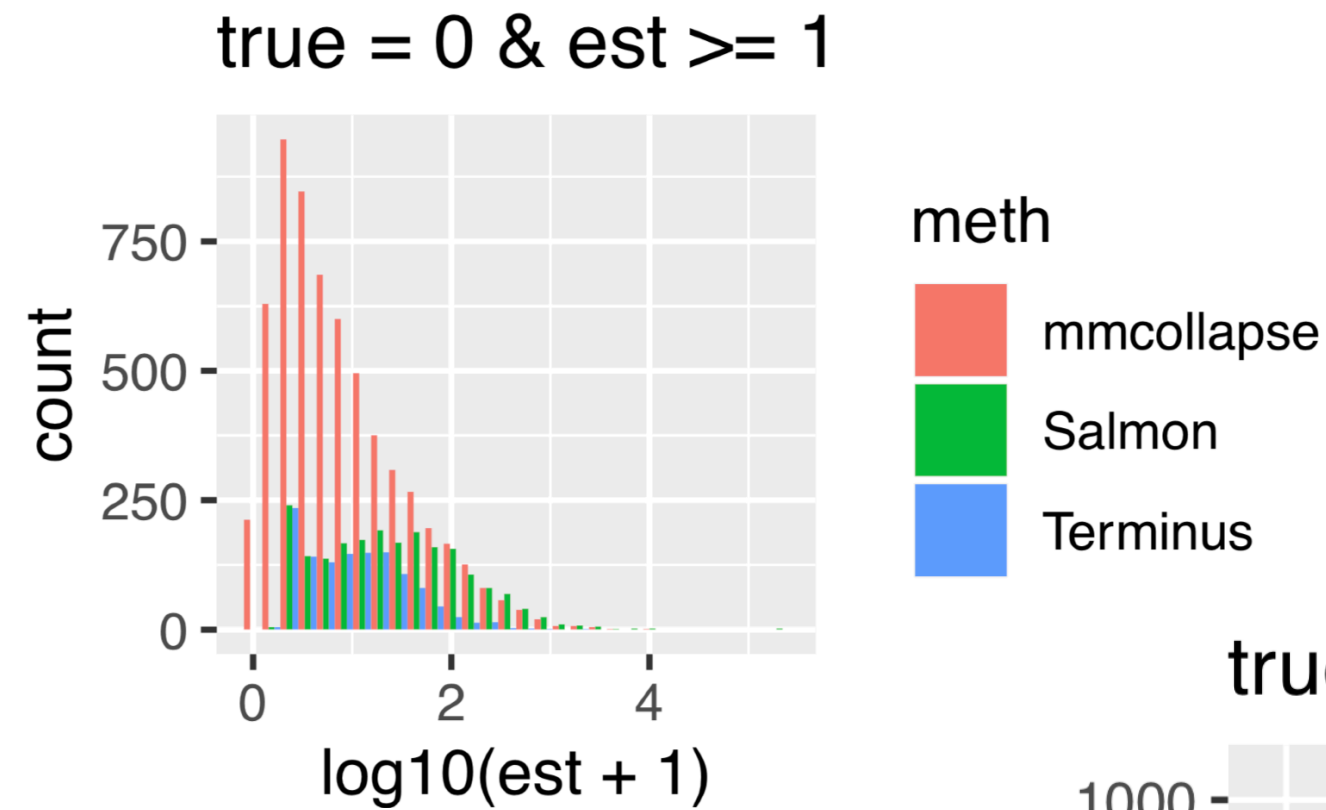
How do terminus groups compare to individual transcripts?



How do terminus groups compare to individual transcripts?



How do terminus groups compare to individual transcripts?



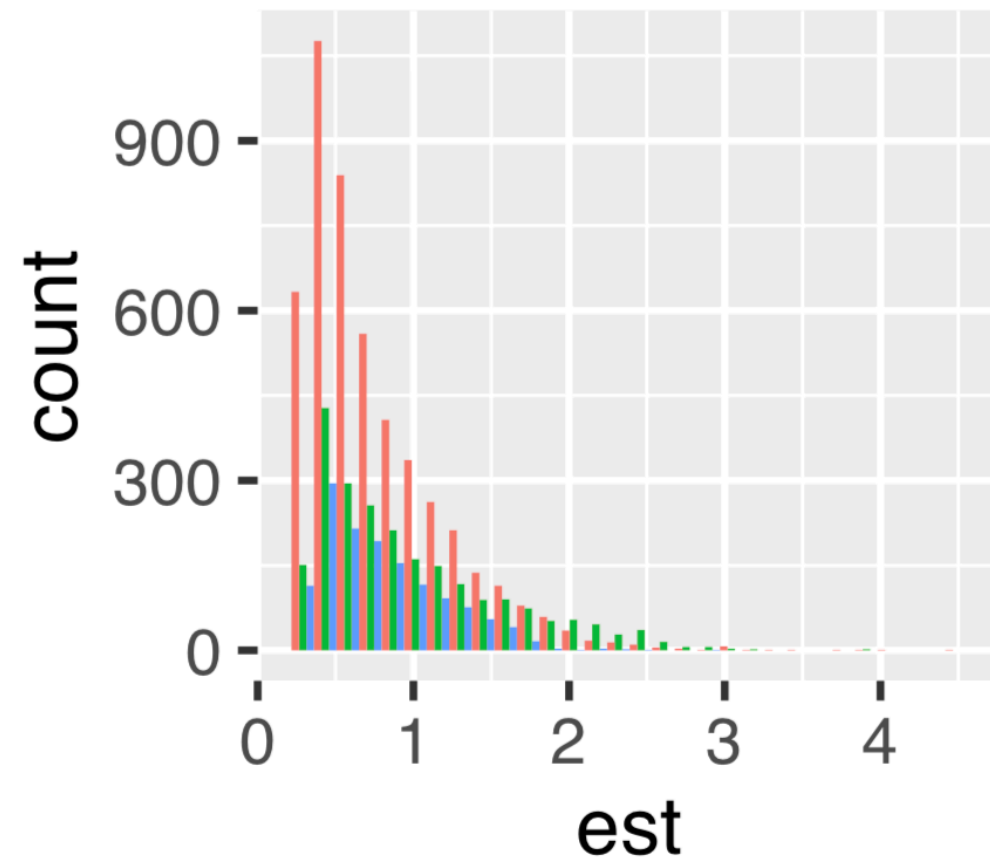
The effect of group when estimating allele-specific expression

Allele-specific expression is *especially* challenging, since the transcripts corresponding to the alleles of a gene are often *nearly identical* (if not actually identical). This exacerbates problems associated with uncertainty!

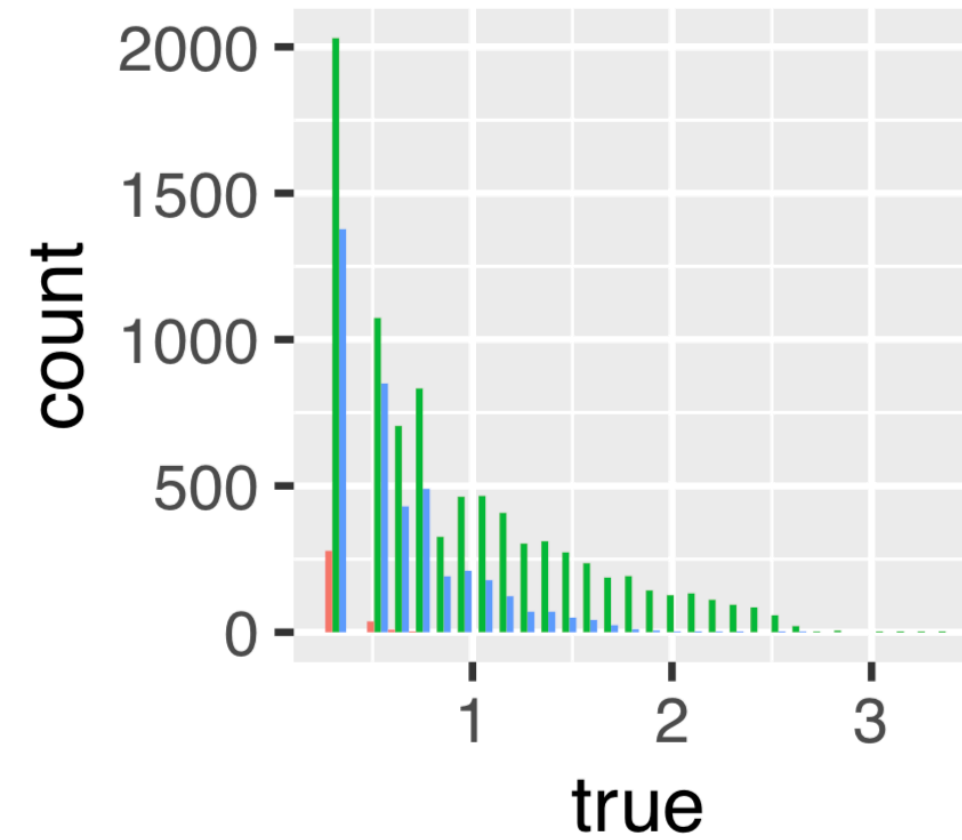
Used the reference mouse transcriptome, annotated variants, and g2gtools* to build a diploid NOD/PWK reference. Quantify transcript abundances *including* all allelic variants.

Allele-specific expression

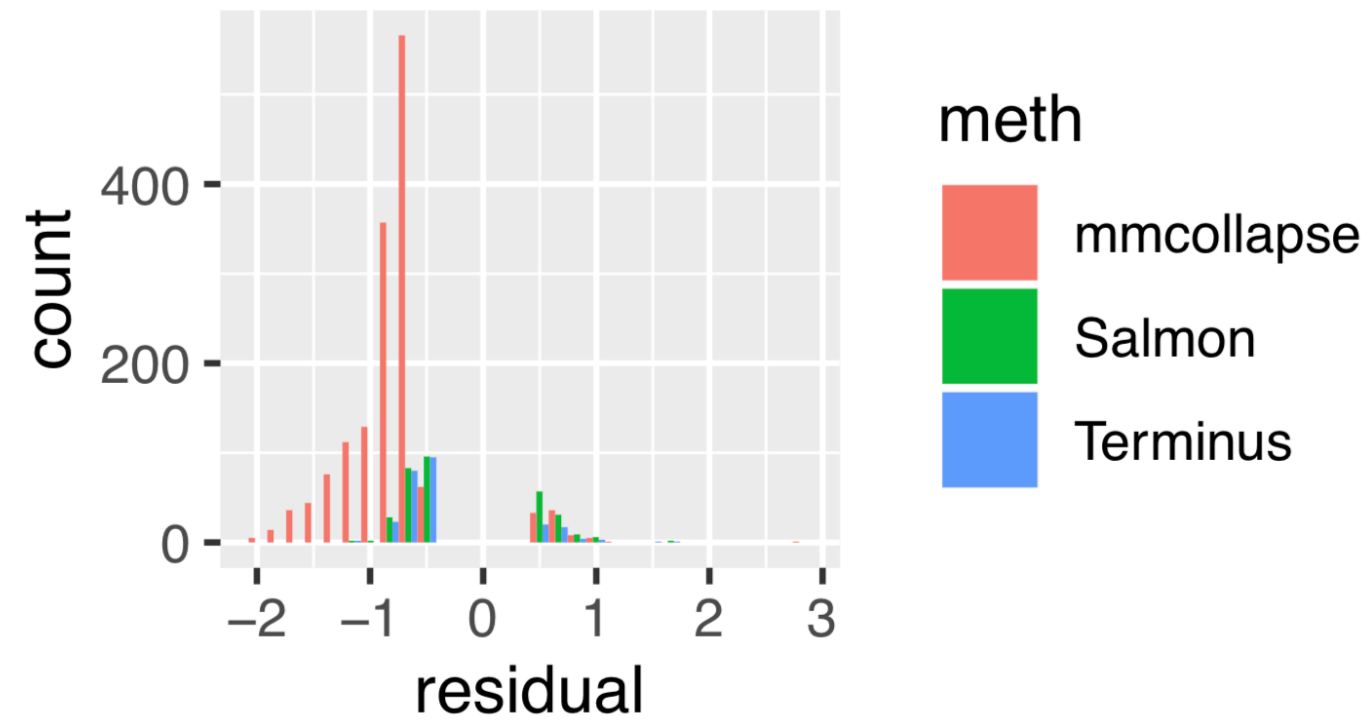
true = 0, est ≥ 1



est = 0



true > 0, est > 0, residual > 0.5



The effect of group when estimating allele-specific expression

Define an *incorrect dominant allele* as a situation where in estimation:

$\text{est}(\text{allele A}) > \text{est}(\text{allele B})$ while $\text{true}(\text{allele B}) > \text{true}(\text{allele A})$

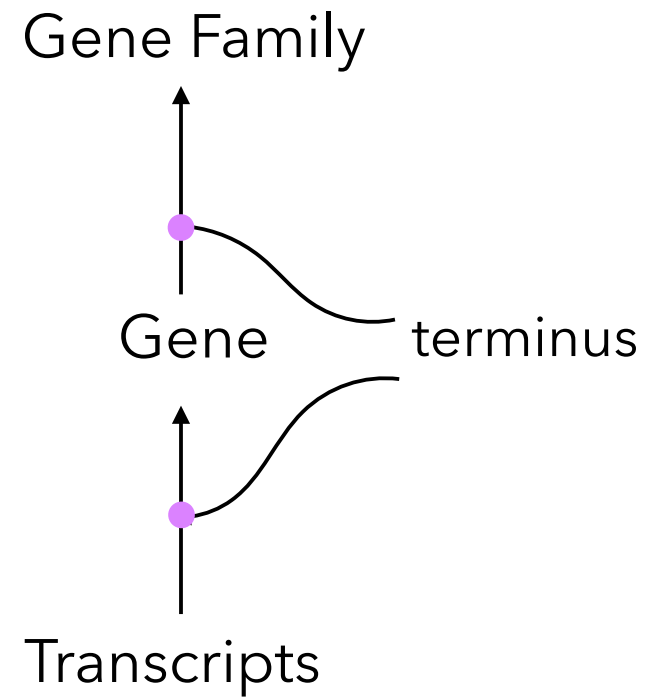
In other words, these are dominant allele switches.

	Salmon	mmcollapse	terminus
Fraction of wrong dominant allele	0.1	0.04	0.04

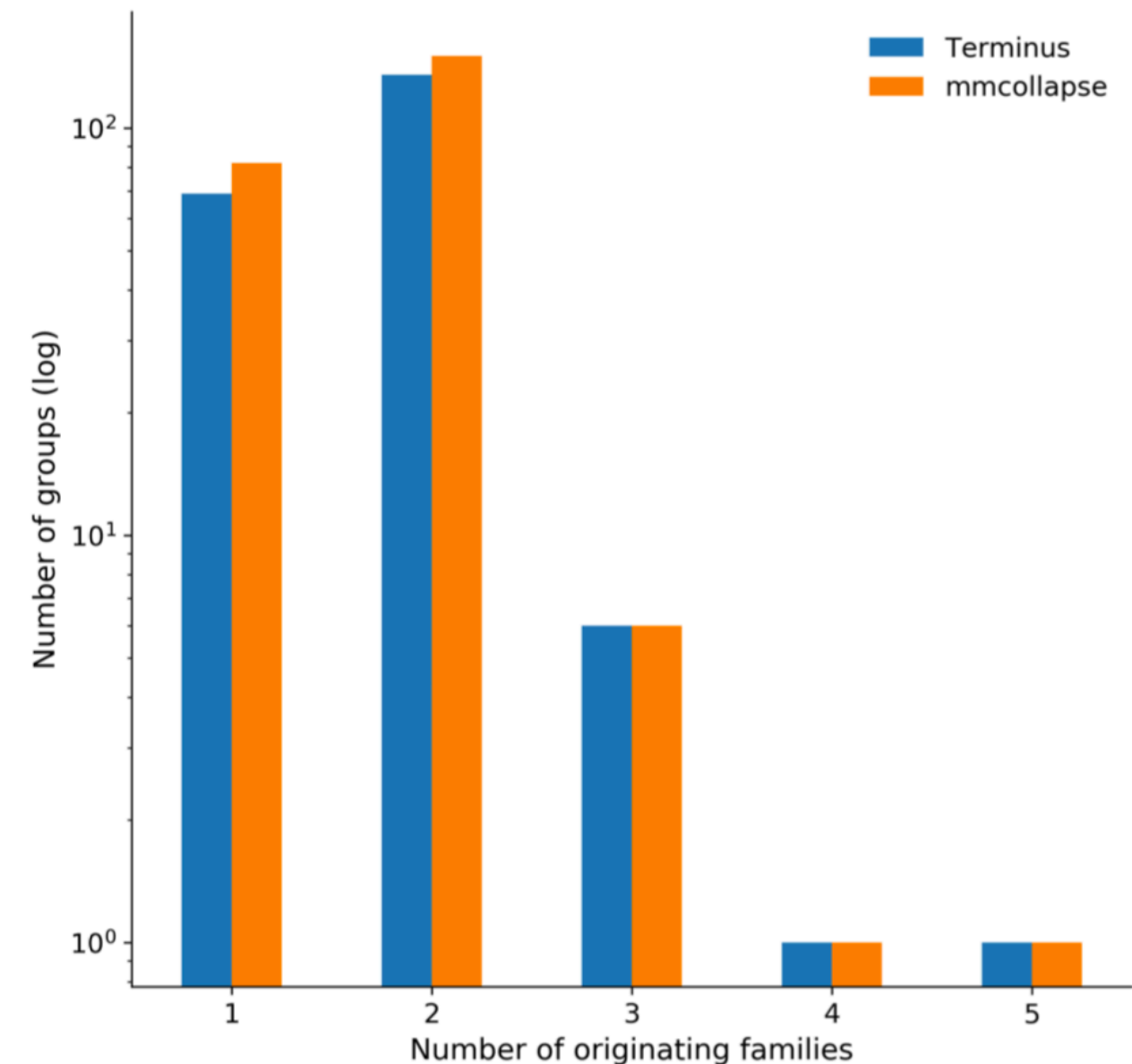
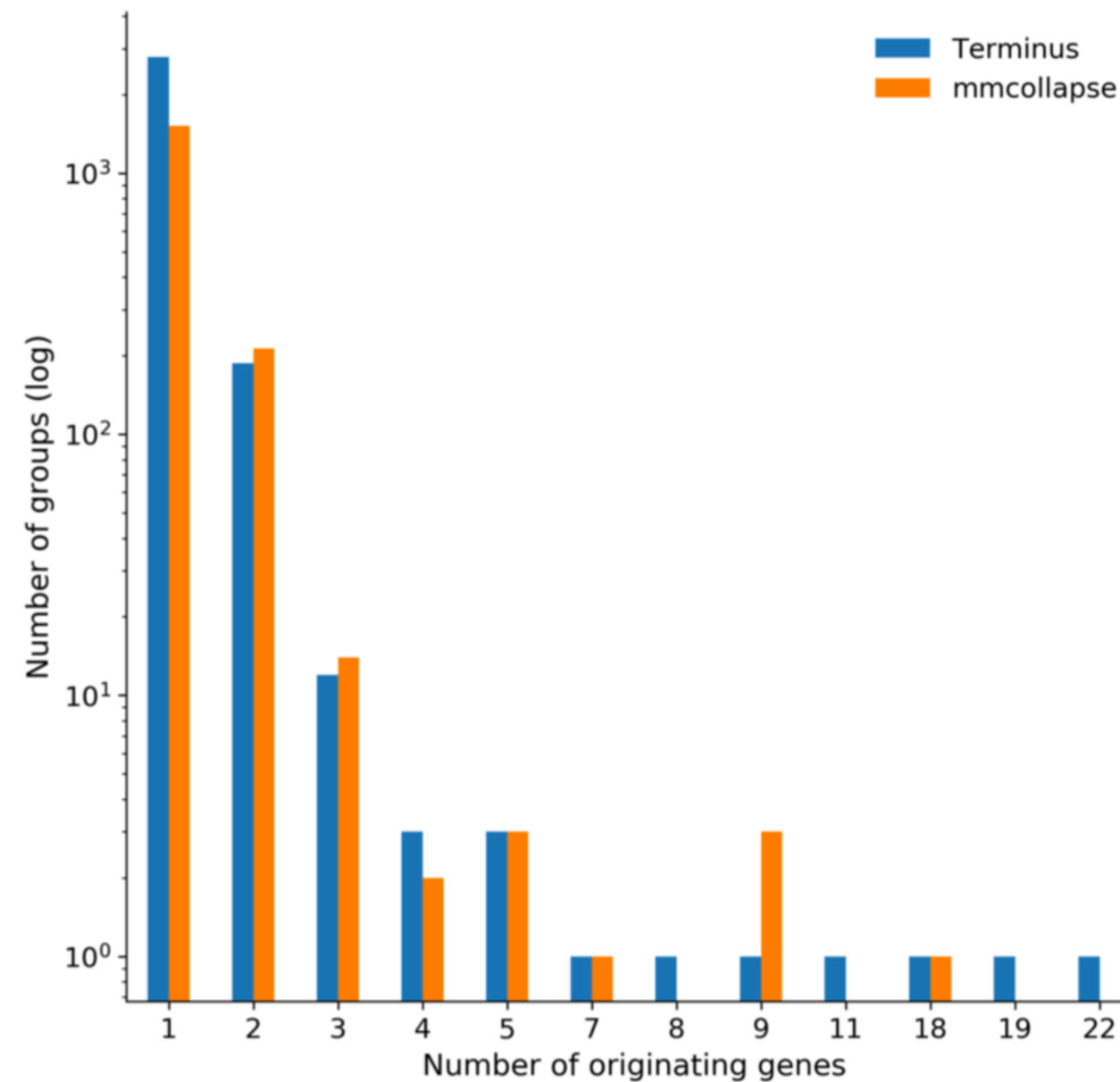
Both mmcollapse & terminus have **fewer than *half* as many wrong dominant alleles.**

Biological plausibility of groups on experimental data

In experimental data, we have no ground truth. But, we can look at how “reasonable” the groups seem.



- Terminus groups are *data-driven*.
- They cannot be replaced via any annotation.
- Though they can span genes, they most often are from one gene or gene family.



Biological plausibility of groups on experimental data

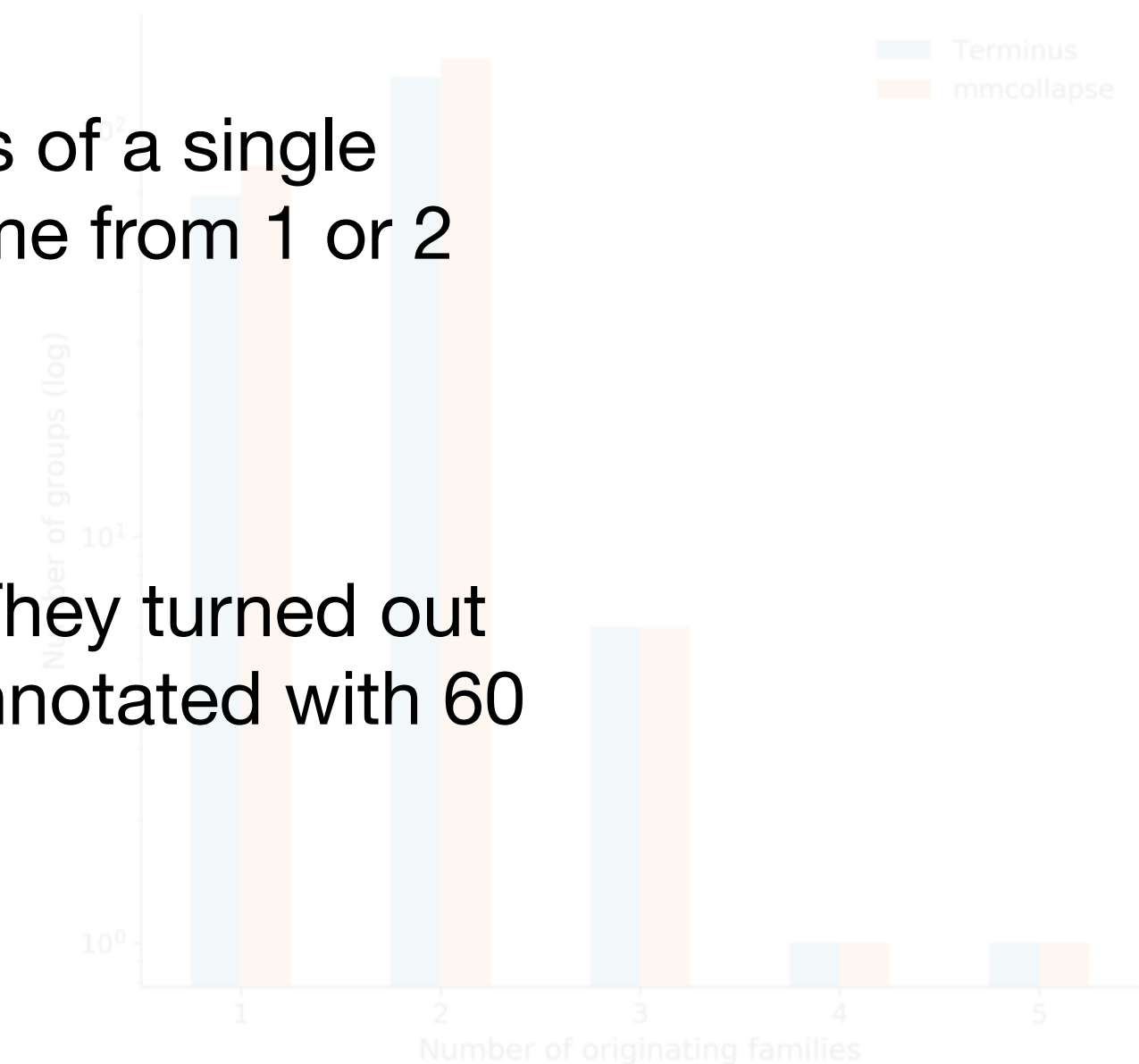
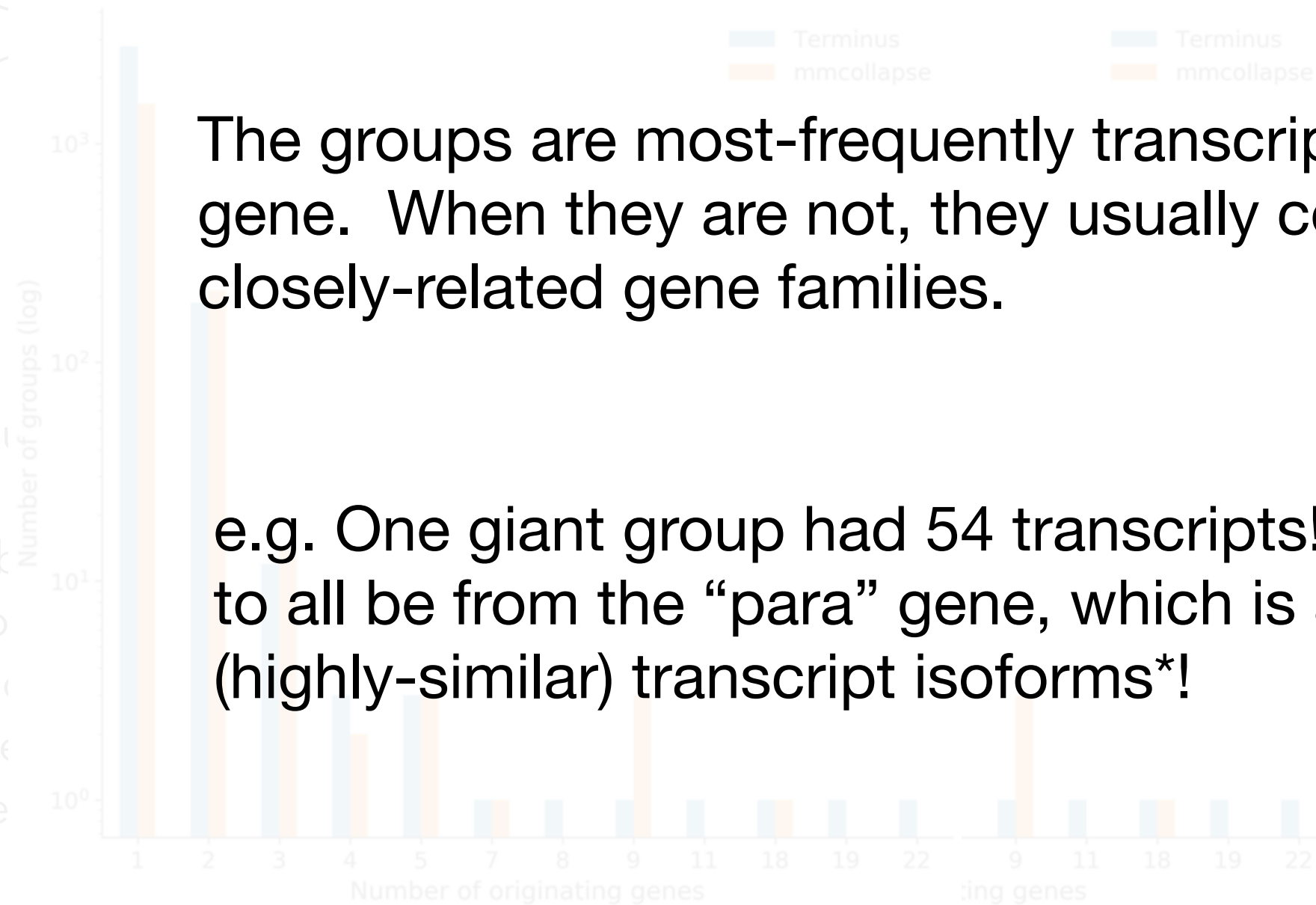
In experimental data, we have no ground truth. But, we can look at how “reasonable” the groups seem.

The groups are most-frequently transcripts of a single gene. When they are not, they usually come from 1 or 2 closely-related gene families.

e.g. One giant group had 54 transcripts! They turned out to all be from the “para” gene, which is annotated with 60 (highly-similar) transcript isoforms*!



- Terminus groups are gene-driven.
- They cannot be assigned any annotation
- Though they are, they most often come from a single gene or gene family



Terminus is fast & memory efficient!

Datasets	Memory(MB)		Time (h:m:s)	
	mmcollapse	Terminus	mmcollapse	Terminus
Human	370,675	2,841	3:46:37	2:04
Mouse	225,457	485	38:55:44	0:12
Drosophila	4,104	310	8:10	0:10

These time and memory profiles are *just* to perform the grouping.

As the annotation complexity and number of expressed transcripts increases, the resource requirements of mmcollapse become untenable. This is a result of the *algorithm*, not the implementation.

Differential Expression Analysis

We have talked about RNA-Seq quantification and how can we quantify and incorporate uncertainty to form transcript groups.

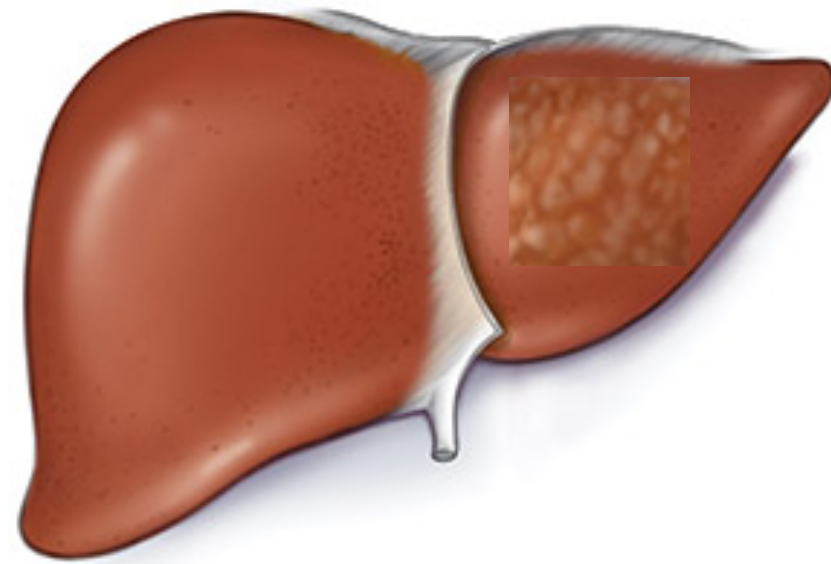
A question can arise is what to do with the quantified transcripts/features.

One most common application is to differential expression analysis

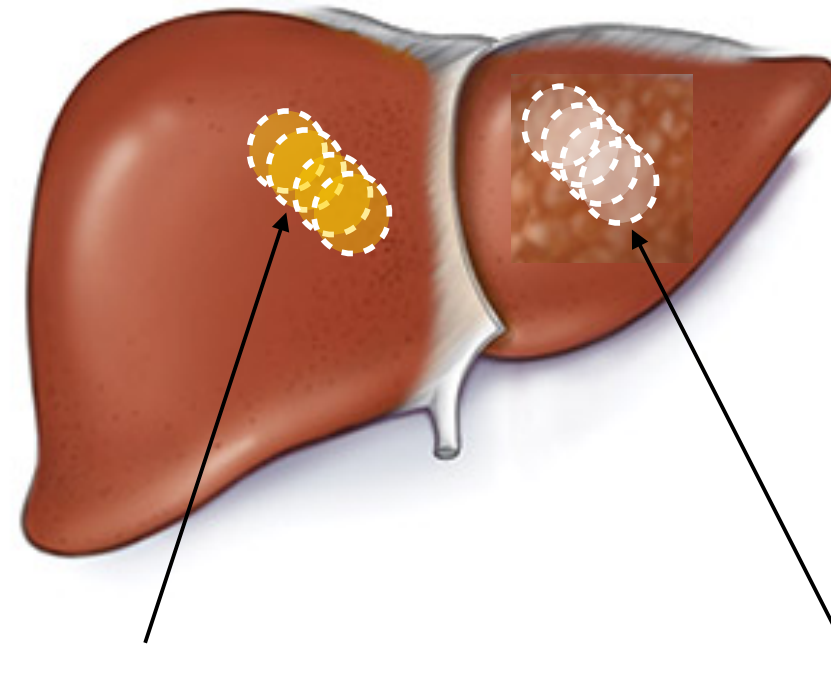
The goal of differential expression analysis is to identify transcripts/genes that are expressed at significantly different levels between two or more experimental conditions (e.g., disease vs. healthy, treated vs. untreated).

By comparing gene expression across conditions, we can gain insights into biological processes affected by the condition(s) of interest.

Case/control differential testing



Case/control differential testing



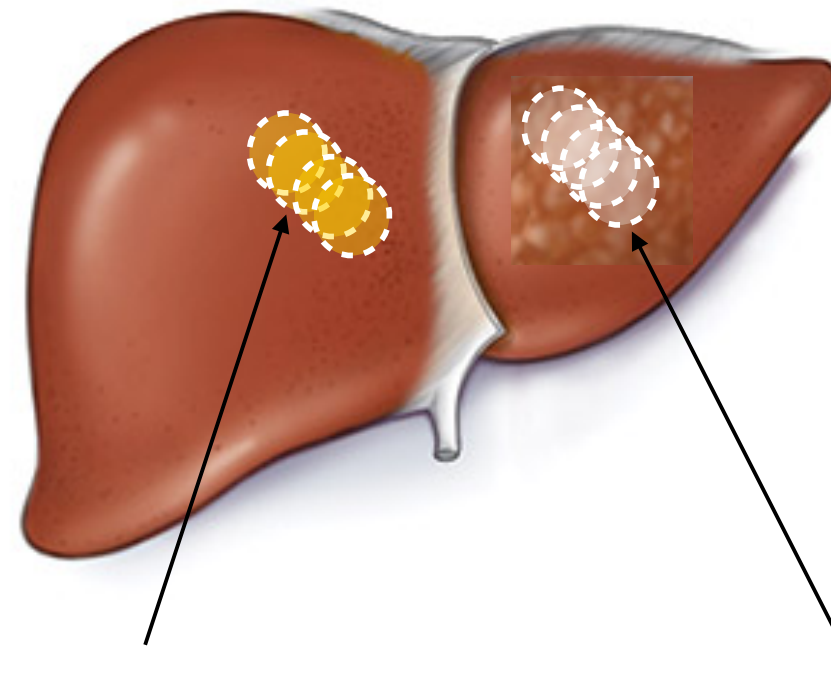
Control — healthy tissue

Case — diseased tissue

What is the difference in gene (transcript) expression profiles between these different samples?

Why multiple samples per “group”?

Case/control differential testing



Control — healthy tissue

Case — diseased tissue

What is the difference in gene (transcript) expression profiles between these different samples?

Why multiple samples per “group”?

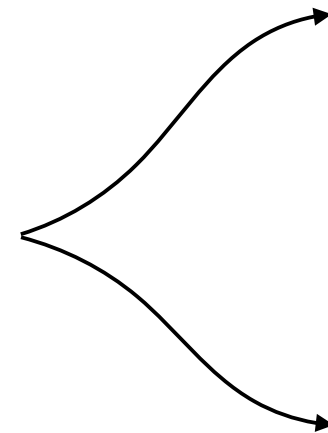
Biological variability! Gene expression naturally varies “within” the same condition, we are mostly interested in expression changes whose magnitude is statistically significant w.r.t this background variability.

Most methods are trying to improve over the (obvious) idea of doing a T-test between conditions (which is actually not too bad when you have a very large number of samples per condition)

Differential expression methods

DGE Tool	Publish Year
edgeR [34]	2010
Cuffdiff/Cuffdiff2 [35, 36]	2012/2013
DESeq2 [37]	2014
limma [38]	2015
DEGseq [39]	2009
baySeq [40]	2010
SAMseq [41]	2013
sleuth [42]	2017
NOIseq [43]	2012
Swish	2019

Non Parametric Methods



Recall Inferential Relative Variance

Inferential relative variance

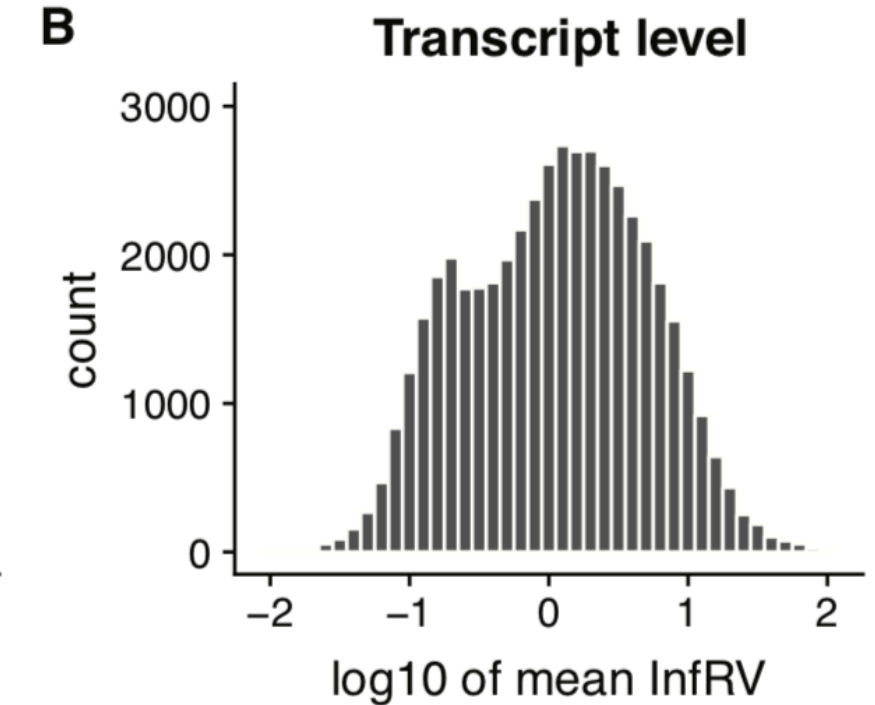
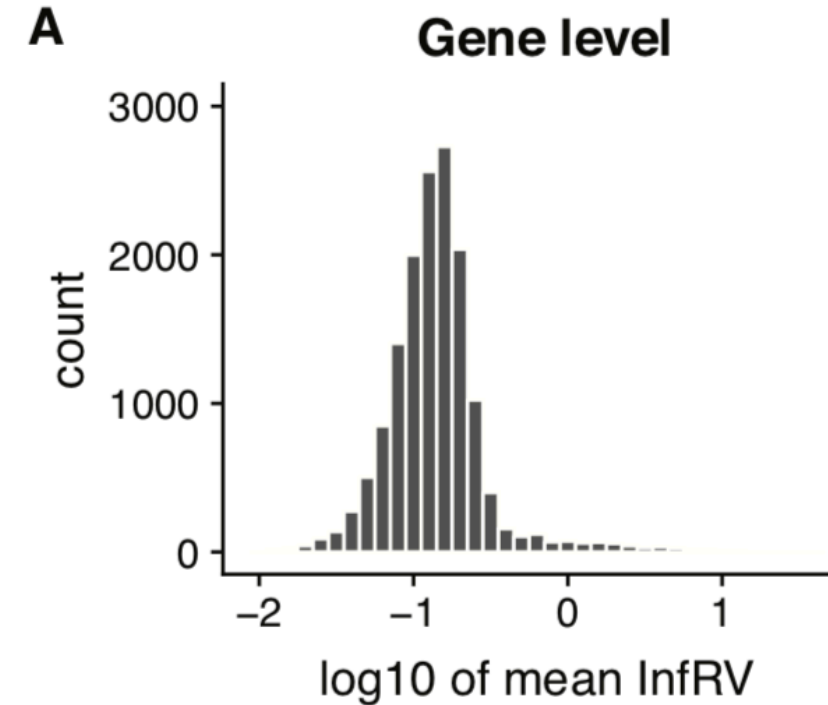
$$\text{infRV}_{\mathbf{g}_i} = \frac{\max(0, \text{var}(\mathbf{g}_i) - \text{mean}(\mathbf{g}_i))}{\text{mean}(\mathbf{g}_i) + p} + s$$

Why (variance - mean) / mean?

Minimally within a sample we have Poisson variance which scales with the mean.

So InfRV = ~0 with no multi-mapping.

Larger the value of InfRV, more uncertainty



Differential expression methods

Take Inferential uncertainty
into account

DGE Tool	Publish Year
edgeR [34] *	2010
Cuffdiff/Cuffdiff2 [35, 36]	2012/2013
DESeq2 [37]	2014
limma [38]	2015
DEGseq [39]	2009
baySeq [40]	2010
SAMseq [41]	2013
sleuth [42]	2017
NOIseq [43]	2012
Swish	2019

*The recent version of edgeR has this functionality

Swish

A non-parametric method that extends upon another non-parametric for RNA-Seq SAMSeq but also includes the inferential replicates

A non-parametric differential testing method does not assume that the underlying data follows a certain probabilistic distribution.

Depending on the nature of the tests, different ways to compute the test-statistic:

- Mann-Whitney U test
- Wilcoxon signed-rank test
- Kruskal-Wallis test
- Permutation test
-

Published online 2 August 2019

*Nucleic Acids Research, 2019, Vol. 47, No. 18 e105
doi: 10.1093/nar/gkz622*

Nonparametric expression analysis using inferential replicate counts

Anqi Zhu¹, Avi Srivastava², Joseph G. Ibrahim¹, Rob Patro² and Michael I. Love^{1,3,*}

¹Department of Biostatistics, University of North Carolina-Chapel Hill, 135 Dauer Drive, Chapel Hill, NC 27599, USA,

²Department of Computer Science, Stony Brook University, Computer Science Building, Engineering Dr, Stony Brook, NY 11794, USA and ³Department of Genetics, University of North Carolina-Chapel Hill, 120 Mason Farm Rd, Chapel Hill, NC 27514, USA

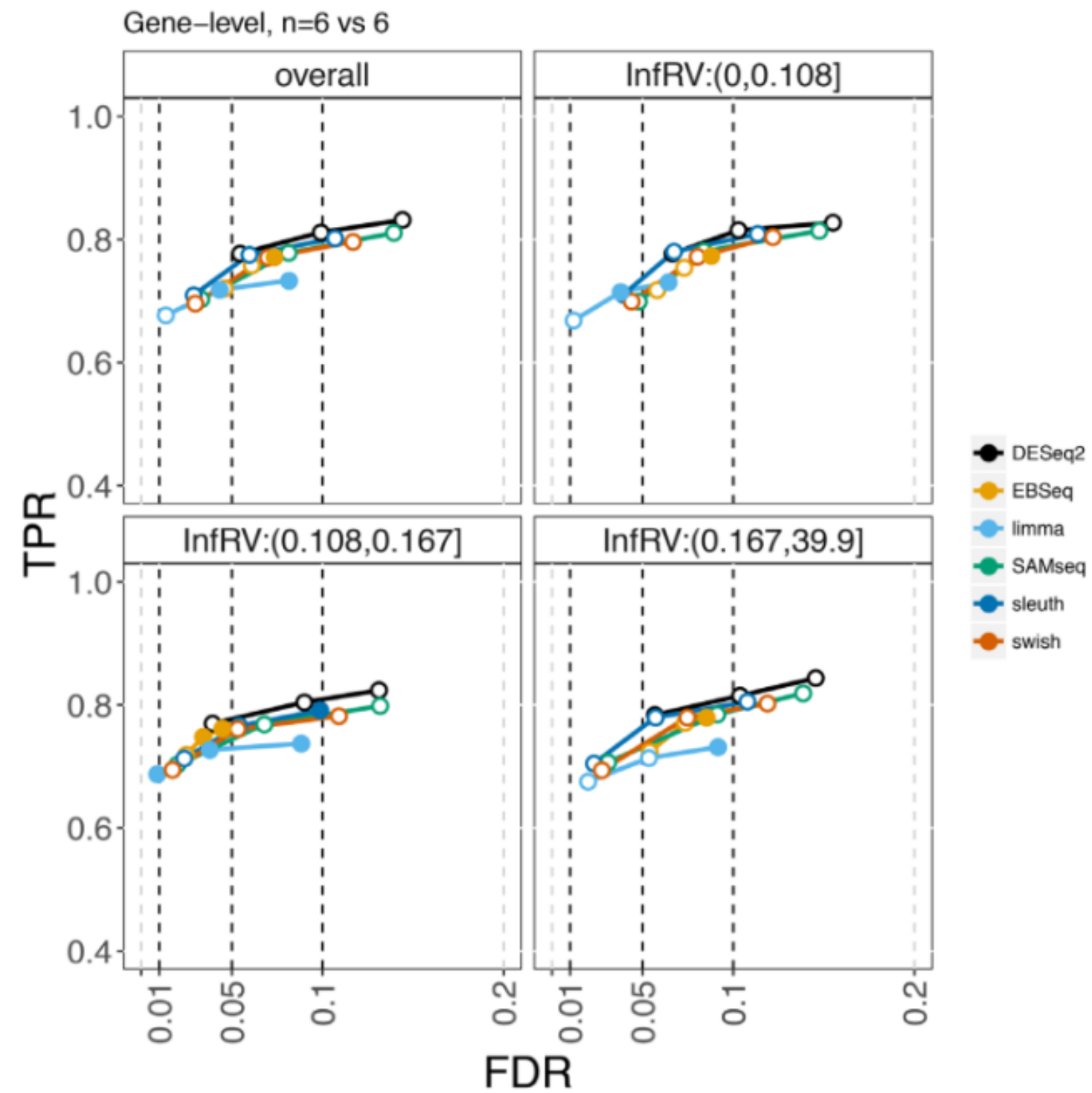
Swish

- Let there be m samples, with m_k denoting samples corresponding to condition $k=1,2$ and Y_{gi}^s be the scaled count from the Gibbs samples ($s = 1..S$).
- If R_{gi}^s denotes the rank for the sample i in m biological replicates for gene g at the s gibbs sample, compute **Mann-Whitney Wilcoxon test statistic**
$$W_g^s = \sum_{i \in C_2} R_{gi}^s - \frac{m_2(m+1)}{2}$$
- Under null hypothesis that gene g is not differentially expressed, W_g^s has expectation zero.
- For each gene g , final test statistic $T_g = \frac{1}{S} \sum_{s=1}^S W_g^s$ is the mean of W calculated over all Gibbs samples.
- This is very similar to SamSeq where mean is computed over downsampled poisson matrices rather than inferential replicates.

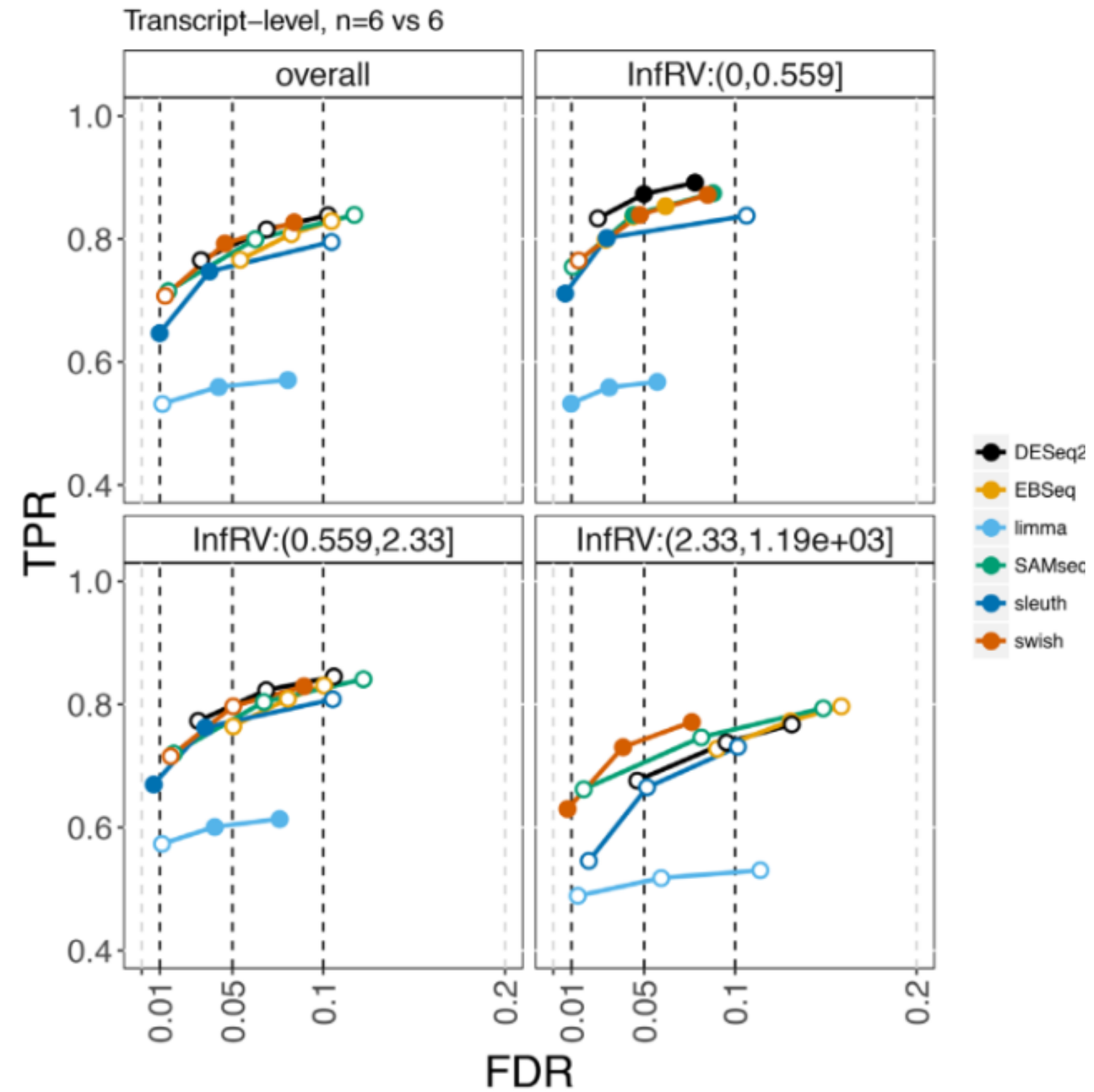
Estimating p-value

- The null distribution for T_g is not known and thus permutation testing is employed to compute the pvalue.
- Permute the samples B times and compute T_g^b based on permuted data (b in $1..B$). p-value is then computed as proportion of how many times T_g is larger than T_i^q (b in $1..G$ and q in $1..B$).

Evaluating DGE



Evaluating DTE



Be careful of parametric assumptions as sample sizes increase

